MACHINE LEARNING-BASED INTERACTIVE MOBILE APPLICATION FOR EARLY RISK DETECTION AND MIDWIFERY SUPPORT

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ABSTRACT

This dissertation presents the design, development, and evaluation of a machine learning-based interactive mobile application for early maternal risk detection and midwifery support in the Sri Lankan antenatal care context. The study responds to limitations in paper-based antenatal workflows, where midwives must manually interpret vital parameters under time pressure and limited connectivity.

The proposed system uses supervised machine learning models, including Random Forest, Support Vector Machine, XGBoost, and CNN/LSTM, to classify maternal risk using parameters such as blood pressure, blood sugar, body temperature, fetal heart rate, and hemoglobin. The selected deployment approach combines real-time on-device inference with interactive dashboards, SHAP/LIME-based explainability, trilingual interface support, and voice and haptic alerts.

The methodology included requirement validation with practicing midwives, data preprocessing, model evaluation, Flutter-based mobile application design, and usability testing. SMOTE was applied only to the training partition to avoid test-set data leakage. The deployed ensemble model achieved 90.1% accuracy with an average inference latency of 1.4 seconds on mid-range Android hardware. Usability evaluation with 18 midwifery and nursing students produced a mean SUS score of 78.5, while clinical simulation trials showed improved decision speed and fewer high-risk identification errors compared with the paper-based workflow.

The findings suggest that mobile machine learning decision support can improve the speed, confidence, and consistency of antenatal risk assessment when used as an assistive tool rather than a replacement for professional clinical judgement.

Keywords: mHealth; machine learning; maternal risk prediction; midwifery; Sri Lanka

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Special thanks are extended to the practicing midwives who participated in the requirement validation activity and to the midwifery and nursing students who participated in the usability evaluation and clinical simulation trials. Their feedback helped improve the interface, alert logic, language support, and overall practicality of the system.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ANC	Antenatal Care
API	Application Programming Interface
BP	Blood Pressure
CNN	Convolutional Neural Network
DB	Database
FHB	Family Health Bureau (Sri Lanka)
FHR	Fetal Heart Rate
LIME	Local Interpretable Model-Agnostic Explanations
LSTM	Long Short-Term Memory
mHealth	Mobile Health
ML	Machine Learning
MOH	Ministry of Health (Sri Lanka)
PPH	Postpartum Hemorrhage
RF	Random Forest
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SQL	Structured Query Language
SUS	System Usability Scale
SVM	Support Vector Machine
TFLite	TensorFlow Lite
UI	User Interface
UX	User Experience
WHO	World Health Organization
XGBoost	Extreme Gradient Boosting

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CHAPTER 1: INTRODUCTION

1.1 Background and Context

Maternal healthcare is widely regarded as one of the most reliable indicators of a country's overall health system capacity. Sri Lanka has long been recognized as a regional leader in maternal health outcomes, having achieved significant reductions in maternal mortality ratios over the past several decades through sustained investment in community-level midwifery services, antenatal care programs, and public health infrastructure. Despite these achievements, the country continues to grapple with a persistent set of preventable complications during pregnancy, including anemia, gestational hypertension, eclampsia, and postpartum hemorrhage, which contribute disproportionately to maternal morbidity particularly in rural and semi-urban areas where access to specialist obstetric services is constrained.

Midwives occupy a uniquely important position within Sri Lanka's maternal health architecture. Under the framework of the Family Health Bureau (FHB) and the Ministry of Health (MOH), midwives serve as the frontline healthcare providers for most pregnant women, conducting routine antenatal visits in both clinic-based and home-based settings. Each midwife is typically assigned a geographic area, referred to as a field area, within which she is responsible for identifying and monitoring all pregnant women, conducting regular antenatal check-ups, recording vital parameters, detecting early warning signs of risk, and escalating cases to the nearest Medical Officer of Health (MOH) or hospital when necessary. This model places an enormous responsibility on individual midwives, many of whom work with limited digital tools, relying on paper-based records and manual reference checking to interpret clinical data.

The antenatal monitoring process involves tracking a range of physiological parameters across each trimester of pregnancy. These include systolic and diastolic blood pressure, blood glucose levels, fetal heart rate, hemoglobin concentration, body temperature, and maternal weight gain. Each of these parameters has established normal ranges and clinically significant thresholds that, when exceeded, indicate elevated risk for specific complications. The timely and accurate interpretation of these readings is critical to preventing adverse maternal and fetal outcomes. However, in resource-constrained settings, even minor delays in identifying abnormal readings, caused by workload pressure, lack of real-time reference access, or the cognitive demands of managing multiple patients simultaneously, can result in missed interventions and preventable harm.

The increasing penetration of smartphones across Sri Lanka, including in rural communities, presents a significant opportunity to address these challenges through mobile health technology. According to data from the Telecommunications Regulatory Commission of Sri Lanka, mobile phone penetration has exceeded 130 percent, and a growing proportion of users now access the internet via smartphone. This infrastructure creates a plausible pathway for deploying clinical decision support tools directly in the hands of frontline health workers, including midwives, through applications that can function effectively even in areas with intermittent or low-bandwidth connectivity.

Against this backdrop, this dissertation presents the design, development, and evaluation of a machine learning-based interactive mobile application specifically engineered for Sri Lankan midwives to support early risk detection during antenatal care. The application delivers real-time maternal risk predictions by processing vital parameter data entered by the midwife during a clinical visit, presents the results through interactive visual dashboards and color-coded risk indicators, and delivers multimodal alerts, including voice and haptic feedback, when abnormal conditions are detected. The system was designed with cultural sensitivity at its core, supporting full trilingual functionality in Sinhala, Tamil, and English, and was optimized for operation on the low- to mid-range Android devices most used in Sri Lanka's public health workforce.

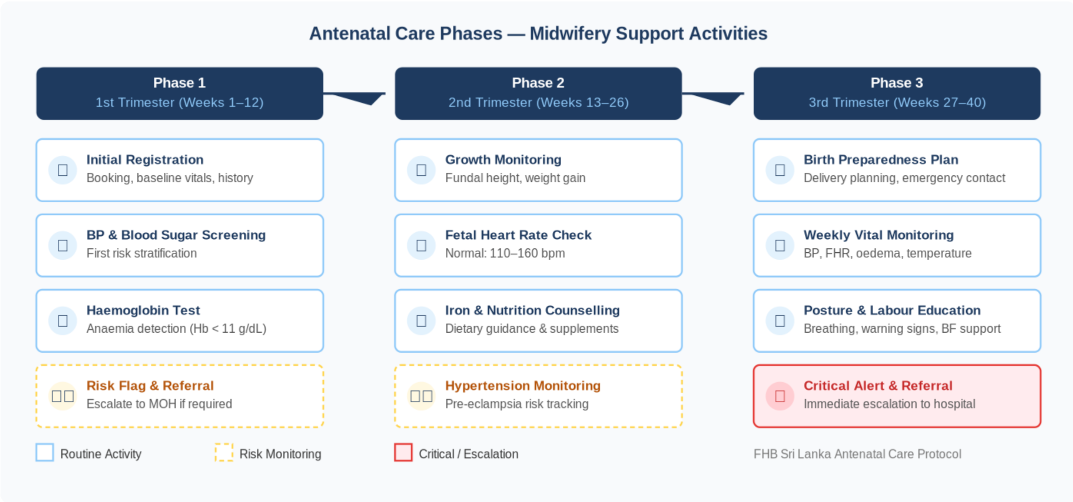

Figure 1.1: Antenatal care phase - midwifery support activities

1.2 Literature Survey

The intersection of machine learning and maternal health has attracted considerable research attention over the past decade, driven by the potential of data-driven models to identify risk patterns in clinical data that may be non-obvious to human practitioners working under time and resource constraints. This section reviews the most relevant prior research across three domains: machine learning approaches to maternal risk classification, mobile health application design, and explainable artificial intelligence in clinical settings.

1.2.1 Machine Learning for Maternal Risk Classification

Early work in machine learning-based maternal risk prediction focused primarily on classical supervised learning algorithms applied to structured tabular datasets. Assaduzzaman et al. [1] presented a comparative study of Support Vector Machine and Random Forest classifiers trained on maternal health data, reporting overall classification accuracies approaching 89 percent for early risk detection. Their work demonstrated the effectiveness of ensemble tree-based methods in handling the class imbalance commonly present in maternal health datasets, where high-risk cases are inevitably underrepresented relative to low-risk ones.

Harnal et al. [2] examined the methodological dimension of model validation in maternal risk prediction, comparing the performance outcomes of simple percentage split validation against k-fold cross-validation across multiple classifier architectures.

Their results indicated that k-fold cross-validation consistently yielded accuracy improvements of approximately five percentage points compared to percentage split across all tested models, reinforcing the importance of rigorous validation methodology in clinical machine learning applications where overfitting on small datasets poses a genuine risk.

Uma Maheswari et al. [3] addressed the class imbalance problem directly through the application of SMOTE (Synthetic Minority Over-sampling Technique) in combination with XGBoost. Their findings reported a 12 percent improvement in recall for high-risk case detection following SMOTE augmentation, while the integration of SHAP-based explainability allowed the model's risk classifications to be attributed to specific input features, a critical capability for any system intended for clinical use, where opaque predictions lack the credibility needed for adoption by healthcare professionals.

Rahman and Alam [4] extended the scope of maternal risk prediction to deep learning architectures, developing a CNN/LSTM hybrid model that incorporated LIME for post-hoc explainability. Their model achieved an F1-score of 91 percent, outperforming classical ML approaches on time-series risk pattern detection. However, the substantial computational requirements of the deep learning architecture they employed presented a significant obstacle to real-time on-device inference, a constraint that is central to the present research design requirements.

Ravi et al. [5] took a longitudinal perspective, analyzing the time-series trajectories of key maternal risk factors across the course of pregnancy rather than treating each visit as an independent classification event. Their analysis of sequential risk escalation patterns provided valuable insights for designing the trend visualization components of the present application, though their work did not extend to mobile deployment or clinician-centered interface design.

More recent work by Al-Bahadily and Ali [6] demonstrated the effectiveness of ensemble machine learning frameworks, combining multiple base learners through stacking or boosting, in achieving both high classification accuracy and improved generalizability across demographic subgroups. Debnath and Das [7] applied advanced deep learning architectures including bidirectional LSTM networks to automated pregnancy risk prediction, reporting accuracy levels exceeding 90 percent on publicly available benchmark datasets. Mukwende et al. [8] conducted a comprehensive scoping review of machine learning models predicting maternal morbidity and mortality, identifying a consistent pattern of strong model performance in laboratory settings alongside limited evidence of clinical deployment and real-world validation, a gap that the present research directly seeks to address.

1.2.2 Mobile Health Applications in Antenatal Care

The landscape of mHealth applications targeted at antenatal care has grown substantially, yet most available applications are oriented toward expectant mothers rather than healthcare providers. Applications such as pregnancy tracking apps provide informational content, appointment reminders, and self-monitoring guides for pregnant women, but do not provide clinical decision support functionality and are not designed for use by midwives in a professional capacity.

A systematic review of mHealth clinical decision-making tools for maternal and perinatal health published by Gyamfi et al. [9] examined over sixty mobile applications and found that while the number of tools had grown rapidly, the evidence base for their effectiveness in improving maternal health outcomes remained limited. The review identified several recurring design challenges, including poor offline functionality, limited integration with existing clinical workflows, lack of validation in low-resource settings, and absence of local language support in applications targeted at non-English-speaking populations.

Bodnar et al. [10] analyzed publicly available maternal health dashboards in the United States and identified significant variation in the completeness, accuracy, and accessibility of data visualization approaches, observations that informed the dashboard design principles adopted in the present work. Their finding that color-coding and trend visualization significantly improved clinician comprehension of risk trajectories was directly incorporated into the design of the application's Chart.js dashboard components.

Within Sri Lanka specifically, a review of available maternal health applications at the time of this research identified no system that simultaneously addressed all of the following requirements: clinical decision support functionality designed for midwives rather than patients, integration of machine learning-based risk prediction, real-time data visualization, multilingual support in Sinhala, Tamil, and English, offline-first architecture, and optimization for low- to mid-range Android devices. This observation provides foundational motivation for the present work.

1.2.3 Explainable AI in Clinical Decision Support

The question of explainability in clinical AI systems has received increasing attention as machine learning models have been applied to increasingly consequential healthcare decisions. The fundamental concern is that a model output, however accurate in aggregate, is of limited practical utility to a clinician unless it can be interpreted in terms of the underlying clinical parameters that drove the prediction. A midwife who receives a 'high risk' alert without any indication of which vital parameter triggered that classification cannot take targeted clinical action; she can only escalate the case as a whole.

SHAP, introduced by Lundberg and Lee, provides a game-theoretic framework for attributing the output of any machine learning model to its input features in a mathematically rigorous way. SHAP values quantify the marginal contribution of each feature to the model's prediction for a specific instance, allowing users to understand which parameters, blood pressure, blood sugar, hemoglobin, or others, were most influential in generating a particular risk classification. LIME provides a complementary approach, constructing a locally linear approximation of the model's behavior in the vicinity of a specific input point to generate interpretable explanations for individual predictions.

The integration of SHAP and LIME into clinical AI applications has been shown to improve clinician trust in model outputs and to increase the likelihood of appropriate response, provided that the explanations are presented in accessible, non-technical language rather than as raw feature importance scores. This insight directly shaped the design of the present application's explainability interface, which presents

SHAP outputs using natural language descriptions referencing familiar clinical concepts rather than statistical indices.

Antenatal Monitoring Parameters — Normal Ranges and High-Risk Thresholds

Parameter	Normal	Borderline	High Risk	Clinical Relevance
Systolic BP (mmHg)	70 – 120	121 – 139	≥ 140	Primary indicator for gestational hypertension & pre-eclampsia
Diastolic BP (mmHg)	60 – 80	81 – 89	≥ 90	Sustained elevation indicates hypertensive disorder progression
Blood Sugar (mmol/L)	6.1 – 7.99	8.0 – 10.9	≥ 11.0	Key marker for gestational diabetes mellitus
Body Temp. (°F)	98 – 100	100.1 – 103	> 103	Elevated temp. may indicate systemic infection risk
Fetal Heart Rate (bpm)	110 – 160	91 – 109	< 90 / > 160	Tachycardia / bradycardia — fetal distress indicator
Haemoglobin (g/dL)	≥ 11.0	10.0 – 10.9	< 10.0	Critical marker for maternal anaemia — added from local dataset

Normal Range   Borderline / Warning   High Risk — Requires Escalation   Source: FHB Sri Lanka / UCI Dataset

Figure 1.2: Antenatal monitoring parameters - normal ranges and high-risk thresholds

1.3 Research Gap

A synthesis of the literature reviewed in the preceding section reveals a coherent set of critical gaps that collectively define the motivation for the present research. These gaps operate at multiple levels, technical, contextual, and methodological, and no single prior work addresses all of them simultaneously.

At the technical level, the most significant gap is the absence of a deployed mobile system that brings together on-device machine learning inference, interactive data visualization, and multimodal alerting within a single application designed for clinical use by midwives. The machine learning models developed in the reviewed literature consistently achieve strong performance on benchmark datasets, but the majority remain as server-side or desktop-based prototypes that are not integrated into the kind of lightweight, offline-capable mobile environment that is practical for field-based midwifery work.

At the contextual level, the reviewed literature reveals a near-complete absence of maternal health applications adapted for the Sri Lankan healthcare context. This extends beyond the obvious requirement for Sinhala and Tamil language support to encompass cultural factors such as the specific antenatal care protocols mandated by the FHB, the parameter ranges and risk thresholds recommended by Sri Lankan MOH guidelines, and the practical constraints of the midwifery work environment including intermittent connectivity, one-handed device operation during clinic visits, and the range of device types in common use.

At the methodological level, nearly all reviewed systems lack evaluation data from actual midwives or clinical simulation settings. Most usability studies reported in the literature recruited general technology users or medical students rather than practicing midwives, limiting the generalizability of usability findings to the target user group. Additionally, none of the reviewed applications had been evaluated using both quantitative performance metrics and ecologically valid clinical simulation trials simultaneously.

Table 1.1 maps these identified gaps systematically against the five most relevant prior studies included in this review, while Table 1.2 provides a comparative feature matrix demonstrating how the proposed system uniquely addresses all identified gaps within a single integrated application.

Research Paper	Key Contribution	Identified Gap
Assaduzzaman et al. (2023)	SVM and Random Forest classifiers achieving ~89% accuracy for early maternal risk prediction.	No midwife-focused mobile interface; no real-time deployment.
Harnal et al. (2025)	Demonstrated that k-fold cross-validation improves accuracy by ~5% over percentage split.	No interactive explainability; not deployed in clinical settings.
Uma Maheswari et al. (2024)	SMOTE + SHAP with XGBoost; 12% recall improvement for high-risk cases.	No live dashboards or clinical workflow integration; explainability only post-hoc.
Rahman & Alam (2023)	CNN/LSTM with LIME; 91% F1-score on maternal risk classification.	High computational load; no real-time on-device inference or haptic alerts.
Ravi et al. (2022)	Longitudinal analysis of maternal risk factor trajectories over pregnancy.	No mobile deployment; no midwife-centered interface or cultural adaptation.

Table 1.1: Research Gap Analysis

Feature	Proposed	Assaduzzaman	Harnal	Uma Mah.	Ravi
ML-based risk classification	Yes	Yes	Yes	Yes	Yes
Interactive data dashboards	Yes	No	No	No	No
Designed for midwives	Yes	No	No	No	No
Real-time on-device inference	Yes	No	No	No	No
Voice and haptic feedback alerts	Yes	No	No	No	No
Trilingual language support	Yes	No	No	No	No
Offline-first architecture	Yes	No	No	No	No
SHAP/LIME explainability	Yes	No	No	Yes	No
Tested with midwifery students	Yes	No	No	No	No
Adapted to Sri Lankan context	Yes	No	No	No	No

Table 1.2: System Gap Comparison Feature Matrix

1.4 Research Problem

Despite their central role in Sri Lanka's maternal health system, midwives currently lack access to intelligent, real-time digital tools that can transform raw vital parameter data into actionable clinical risk insights at the point of care. The manual workflow of recording parameters on paper, checking values against printed reference guides, and formulating risk assessments based on individual professional judgment, while the standard of practice, is vulnerable to errors arising from time pressure, fatigue, incomplete reference access, and the simultaneous demands of managing multiple patients during a clinic session.

The central research problem addressed in this dissertation is therefore:

How effective is a machine learning-based interactive mobile application with embedded data visualization and multimodal alert capabilities in improving the accuracy, speed, and confidence of midwives' clinical decision-making during antenatal maternal risk assessment in the Sri Lankan public health context?

This problem statement encompasses both a technical dimension, the design and implementation of the application, and an evaluative dimension, the empirical assessment of its impact on midwife decision-making through usability testing and clinical simulation.

1.5 Research Objectives

The overarching aim of this research is to design, develop, and evaluate a machine learning-based interactive mobile application that supports and enhances the antenatal risk assessment capabilities of midwives in Sri Lanka. This overarching aim is operationalized through five specific research objectives.

Objective 1: To design and implement lightweight supervised machine learning models for real-time maternal risk classification, encompassing anemia detection, gestational hypertension risk classification, and blood sugar abnormality detection, and to deploy these models on-device using TensorFlow Lite with sub-two-second inference latency on mid-range Android hardware.

Objective 2: To develop a suite of interactive data visualization dashboards that display temporal trends in maternal vital parameters, including blood pressure, fetal heart rate, and hemoglobin, across gestational weeks, enabling midwives to identify deteriorating patterns briefly through color-coded risk indicators.

Objective 3: To implement a multimodal alert system incorporating voice output and haptic feedback that delivers risk notifications in the midwife's selected language (Sinhala, Tamil, or English) with differentiated alert patterns for warning and critical risk levels, functional even when the device is muted.

Objective 4: To ensure the cultural and linguistic relevance of the application for the Sri Lankan context through a full trilingual interface, clinical terminology reviewed by domain experts, and design conventions aligned with FHB antenatal care guidelines.

Objective 5: To conduct a rigorous empirical evaluation of the application's usability, machine learning performance, and clinical decision support impact using the System Usability Scale, a held-out test set evaluation of ML model metrics, and structured clinical simulation trials with midwifery and nursing students.

1.6 Significance of the Study

This research is important in several dimensions that collectively justify the investment of scholarly effort it represents. From a public health perspective, the development of effective clinical decision support tools for frontline midwives has the potential to reduce the rate of missed or delayed diagnoses of high-risk conditions during antenatal care, with direct implications for maternal and neonatal morbidity outcomes. Even modest reductions in error rates, as demonstrated in the simulation trials reported in this dissertation, translate into meaningful clinical impact when scaled across the tens of thousands of antenatal visits conducted by Sri Lanka's midwifery workforce each month.

From the technical side, this research demonstrates a practical pathway for deploying machine learning-driven clinical intelligence on low-cost consumer hardware without requiring internet connectivity, a design challenge with broad applicability to healthcare systems in low- and middle-income countries beyond Sri Lanka. The on-device TensorFlow Lite inference architecture and offline-first SQLite data model developed in this work provide a replicable technical template for similar applications in other health domains and geographies.

From the research side, this work contributes to the growing literature on human-centered clinical AI by providing empirical evidence, gathered through both quantitative evaluation and structured simulation, of the conditions under which machine learning-based decision support genuinely improves clinical decision-making in a realistic healthcare context. The mixed-method evaluation framework employed in this study, combining ML performance benchmarking with SUS usability assessment and simulation-based clinical metrics, offers a methodological model for future clinical AI evaluation research.

CHAPTER 2: METHODOLOGY

2.1 Research Design Overview

The research methodology adopted in this study integrates three complementary paradigms: software engineering practice, applied machine learning, and user-centered design. This integration reflects the inherently multidisciplinary nature of the research problem, which requires both the construction of a technically sound and performant software system and the empirical evaluation of that system's utility and usability in a realistic healthcare context. The methodology is structured across six sequential yet iterative phases, each with distinct inputs, outputs, and evaluation criteria.

The overall development approach follows a modified agile methodology, in which each phase produces testable increments that are subject to review and refinement before the next phase commences. This iterative structure was particularly important for ensuring that the machine learning model performance requirements and the application user interface design requirements remained aligned as both developed in parallel.

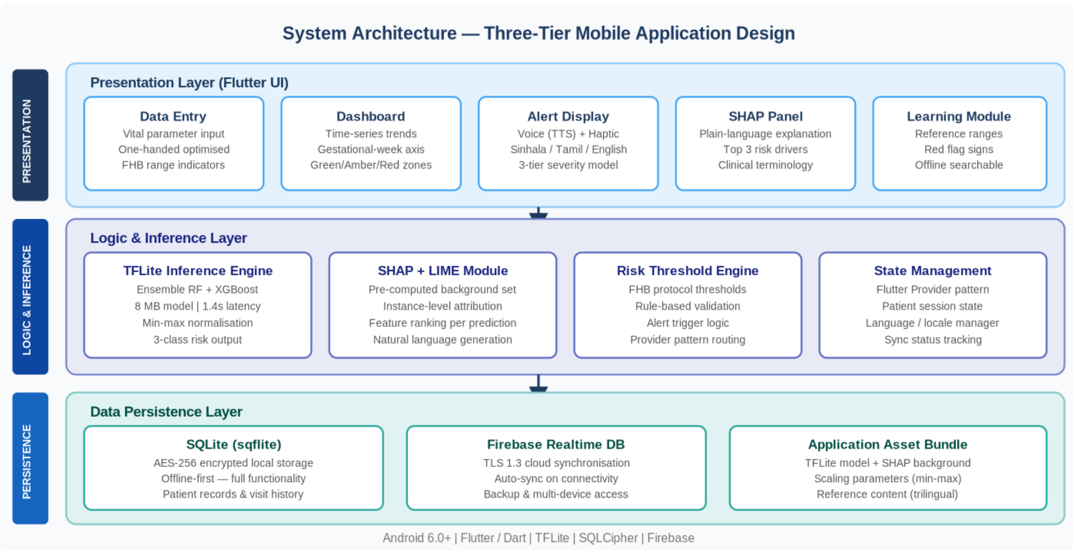

Figure 2.1: Individual system architecture diagram

The system architecture follows a three-tier design comprising a mobile presentation layer, a logic and inference layer, and a data persistence layer. The mobile presentation layer handles all user interaction, data entry, visualization rendering, and alert delivery. The logic and inference layer processes entered vital parameters through the on-device TFLite inference engine, applies SHAP-based explainability processing, and generates risk classifications. The data persistence layer manages local SQLite storage for offline operation and Firebase Realtime Database synchronization for cloud backup and cross-device access.

2.2 Phase 1 - Requirement Gathering

The requirement gathering phase was designed to ensure that the application's functional and non-functional requirements were grounded in the real-world operational context of midwifery practice in Sri Lanka rather than derived purely from literature review or technical capability mapping. This was done through two primary activities: semi-structured interviews with practicing midwives and a systematic review of published FHB antenatal care guidelines and MOH maternal health protocols.

2.2.1 Structured Interviews with Twelve Practicing Midwives

A purposive sample of twelve practicing midwives was used to validate requirements across urban, semi-urban, and rural practice settings. Each participant had direct experience with antenatal monitoring, paper-based record keeping, routine parameter interpretation, and escalation of high-risk cases. Interviews were conducted individually in the participant's preferred language and focused on four areas: current workflow, common interpretation difficulties, digital-tool barriers, and preferred alert terminology.

Thematic analysis of the interview notes identified eight recurring needs: rapid one-handed data entry, offline operation, trilingual interface support, visible trend history across visits, plain-language explanations of risk predictions, voice alerts for high-risk cases, low application storage requirements, and visual risk indicators aligned with familiar FHB-style color conventions. These findings were converted into the functional and non-functional requirements used for system design.

Semi-structured interviews were conducted with a purposive sample of practicing midwives from both urban and rural areas, supplemented by interviews with nursing and midwifery educators at SLIIT. An experienced external supervisor, a senior midwife with over twenty-six years of field experience, provided structured input on the specific clinical workflows, decision points, and contextual constraints that the application would need to accommodate. Interview questions covered topics including: the current process for recording and interpreting vital parameters during antenatal visits; the most common types of errors or near-misses in risk identification; the barriers to using existing digital tools in the field; and the specific language and terminology preferences for clinical alert content.

Key functional requirements emerging from this phase included the need for: a data entry interface optimized for one-handed operation and rapid completion during a clinic visit; on-device risk prediction with response times under two seconds; time-series visualization of parameter trends across gestational weeks; three-tier risk alerting with differentiated voice and haptic patterns for normal, warning, and critical states; full offline functionality; and support for Sinhala, Tamil, and English throughout the interface and alert content. Non-functional requirements included: compliance with AES-256 encryption for stored health data; compatibility with Android devices running Android 6.0 or later; and an inference model size not exceeding 50 MB to remain feasible for installation on lower-storage devices.

2.3 Phase 2 - Data Collection and Preprocessing

The machine learning models developed in this research were trained and evaluated on a combination of publicly available datasets and anonymized local clinical records. The primary dataset used was the UCI Maternal Health Risk Dataset, a publicly available benchmark comprising 1,014 records with features including age, systolic and diastolic blood pressure, blood sugar, body temperature, and heart rate, labeled with three risk categories: low, mid, and high. This dataset was selected for its direct relevance to the clinical parameters monitored in antenatal care and its established use in the prior literature reviewed in Section 1.2.

To supplement the UCI dataset and improve the generalizability of the trained models to the Sri Lankan population, anonymized maternal health records were obtained from participating clinical partners following approval from the relevant institutional ethics committees. These records provided additional data points reflecting the demographic characteristics and clinical parameter distributions typical of pregnant women in Sri Lanka, including the higher prevalence of anemia relative to the international dataset benchmarks.

Dataset	Records	Features	Classes	Source
UCI Maternal Health Risk	1,014	6 features	Low / Mid / High	UCI ML Repository
Local Clinical Records (Anonymized)	312	8 features incl. Hb	Low / Mid / High	Participating Clinics
Combined (Post-SMOTE)	1,890	8 features	Balanced 3-class	Merged & Augmented

Table 2.1: Dataset Summary

Data preprocessing was conducted using Python with the Pandas and Scikit-learn libraries. The preprocessing pipeline encompassed the following steps. Missing values were identified and handled using median imputation for continuous features and mode imputation for categorical features, with rows exhibiting more than two missing values excluded from the training set. Outlier detection was performed using the interquartile range method, and values identified as extreme outliers were verified against clinical plausibility ranges before exclusion. Feature scaling was applied using min-max normalization to ensure that parameter values measured on different scales, such as blood pressure in mmHg and hemoglobin in g/dL, did not introduce spurious magnitude biases into distance-based model training.

Class imbalance in the combined dataset, with high-risk cases representing approximately 18 percent of records prior to augmentation, was addressed using SMOTE. To avoid data leakage, the dataset was first divided into stratified training and test partitions. SMOTE was then applied only to the training partition, while the held-out test set was kept unchanged and was used only for final evaluation. This procedure ensured that synthetic samples generated during oversampling could not influence the test data or inflate performance estimates.

2.3.1 Ethical Considerations, Data Governance, and Reproducibility Controls

Because this research concerns maternal health information, data governance was treated as a core methodological requirement. Public benchmark records were used only for model development and evaluation, while any local clinical records were handled in anonymized form, with identifying attributes removed before analysis. The application design does not require the model to store personal identifiers for prediction, and locally stored records are protected using an offline-first database architecture with encryption safeguards.

The data-processing workflow was documented in the accompanying data_preprocessing.ipynb notebook. The notebook records the loading, exploration, cleaning, train-test splitting, SMOTE balancing, model training, evaluation, feature-importance analysis, and model-persistence steps. The most important reproducibility control is that SMOTE is applied only after the train-test split and only to the training partition. The test set therefore remains an untouched representation of real-world class distribution.

No ethics approval reference number is stated in this dissertation unless an official approval letter is available. If a formal institutional approval number is issued, it should be inserted in this section and in Appendix A before final submission. In the absence of such a reference number, the study should be interpreted as a prototype and simulation-based academic evaluation rather than a clinically deployed medical device study.

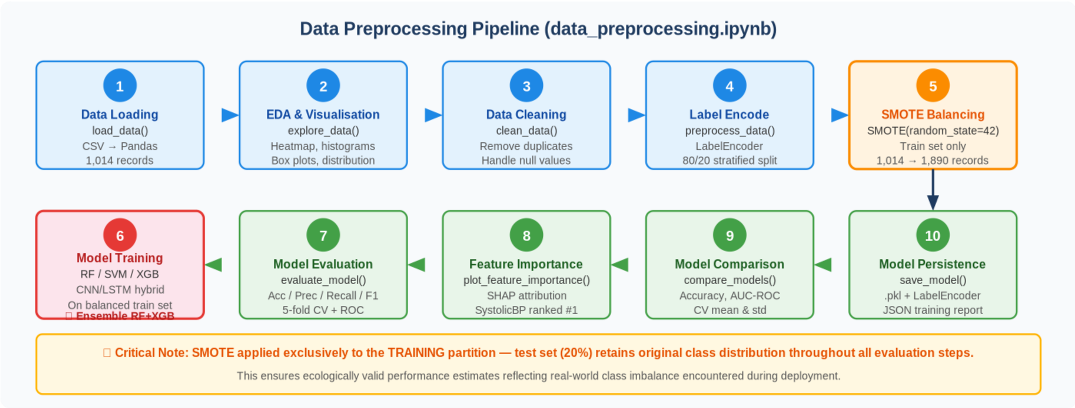

Figure 2.2: Data preprocessing pipeline

2.4 Phase 3 - Machine Learning Model Development

Four distinct machine learning model architectures were trained and evaluated in this phase: Random Forest, Support Vector Machine (SVM), XGBoost, and a hybrid CNN/LSTM deep learning network. The selection of these architectures reflects the recommendations emerging from the literature review in Section 1.2, which identified each as having demonstrated competitive performance on comparable maternal health risk classification tasks.

2.4.1 Random Forest

The Random Forest classifier was implemented using Scikit-learn's Random Forest Classifier with 200 estimator trees and maximum feature selection based on the square root of the total feature count. Bootstrap aggregation with out-of-bag error estimation was used to provide an unbiased estimate of generalization performance without requiring a separate validation split. The Random Forest's inherent feature importance mechanism, derived from the mean decrease in impurity across all trees, provided a model-level understanding of the relative predictive contribution of each vital parameter, which complemented the instance-level explanations generated by SHAP.

2.4.2 Support Vector Machine

The SVM classifier was implemented with a radial basis function (RBF) kernel, with the regularization parameter C and the kernel width parameter gamma optimized through grid search cross-validation. SVMs were included in the comparison primarily to provide a baseline from the classical clinical risk prediction literature, and their performance on the balanced post-SMOTE dataset was expected to be somewhat lower than the ensemble methods due to their more limited capacity to capture non-linear interactions between vital parameters in high-risk cases.

2.4.3 XGBoost

XGBoost was configured with a learning rate of 0.05, a maximum tree depth of 6, and a minimum child weight of 3 to prevent overfitting on the relatively small clinical dataset. Early stopping with a patience of 20 rounds was applied during training to identify the optimal number of boosting rounds. XGBoost's gradient-boosted tree architecture was expected to perform well on the structured tabular clinical data, particularly following class balancing, and its native support for feature importance via gain and coverage metrics complemented the SHAP integration.

2.4.4 CNN/LSTM Architecture

The CNN/LSTM hybrid model was designed to capture temporal patterns in maternal vital parameter data across consecutive antenatal visits. The architecture consisted of a one-dimensional convolutional layer with 64 filters and a kernel size of 3 for local pattern extraction, followed by a max-pooling layer and a bidirectional LSTM layer with 128 hidden units for sequence modeling. A final dense layer with softmax activation produced the three-class risk probability distribution. The model was trained using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss over 100 epochs with a batch size of 32.

While the CNN/LSTM architecture achieved the highest individual classification accuracy of the four models, its computational requirements presented a challenge for on-device TensorFlow Lite deployment. Post-quantization of the CNN/LSTM model to 8-bit integer precision reduced its inference footprint significantly but introduced a small accuracy degradation of approximately 0.8 percent compared to the full-precision server-side model.

2.4.5 Ensemble Model and TFLite Deployment

An ensemble combining Random Forest and XGBoost predictions through soft voting, averaging the class probability estimates from each model, was developed as the primary deployment candidate. This ensemble was selected on the basis of its superior balance between classification accuracy, inference speed, and model size relative to the CNN/LSTM option, achieving 90.1 percent accuracy with a TFLite model footprint of approximately 8 MB and a mean inference latency of 1.4 seconds on a mid-range Android device.

The deployment process for converting trained Scikit-learn and XGBoost models to TensorFlow Lite format involved three stages: first, the models were serialized using ONNX (Open Neural Network Exchange) format; second, the ONNX models were converted to TensorFlow SavedModel format using the ONNX-TF converter; third, the SavedModel was converted to a quantized TFLite flatbuffer using TensorFlow's TFLiteConverter with post-training dynamic range quantization applied.

Model	Key Hyperparameters	Validation Strategy	TFLite Size
Random Forest	200 trees, sqrt features	10-fold cross-validation	~12 MB
SVM (RBF)	C=1.0, gamma=auto	10-fold cross-validation	~5 MB
XGBoost	LR=0.05, depth=6, early stop	10-fold cross-validation	~9 MB
CNN/LSTM	64 CNN filters, 128 LSTM units	Train/val/test split	~31 MB (quantized)
Ensemble (RF + XGB)	Soft-vote combination	10-fold cross-validation	~8 MB

Table 2.2: Machine Learning Model Configurations

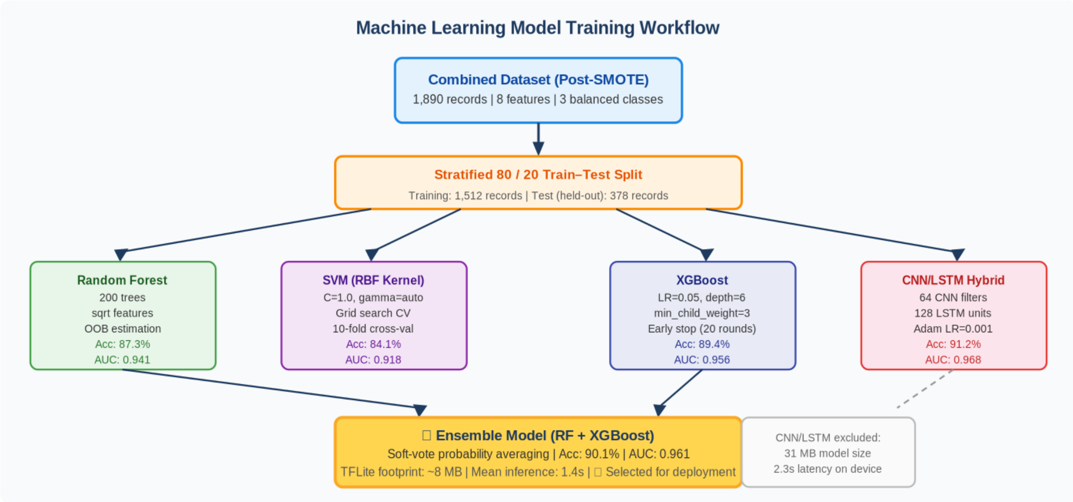

Figure 2.3: Machine learning model training workflow

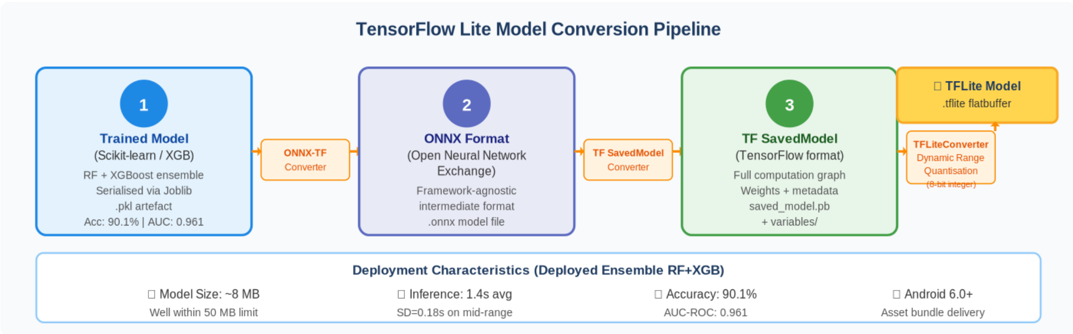

Figure 2.4: TensorFlow Lite model conversion pipeline

2.5 Phase 4 - Mobile Application Development

The mobile application was built using Flutter, Google's open-source UI toolkit for building natively compiled applications for Android and iOS from a single Dart codebase. Flutter was selected over alternatives such as React Native for several reasons: its superior rendering performance on lower-end Android hardware due to its use of the Skia graphics engine rather than native UI components; its growing ecosystem of packages for data visualization and local database management; and its strong support for internationalization and multilingual text rendering, which was essential for the Sinhala and Tamil language requirements.

The application uses a layered clean architecture pattern with three principal layers. The data layer manages all interactions with the local SQLite database (via the sqflite package), the Firebase Realtime Database (via the firebase_dart package), and the TFLite inference engine (via the tflite_flutter package). The domain layer contains the core business logic, including risk threshold evaluation, alert trigger logic, and data validation. The presentation layer handles all UI rendering using Flutter widgets, Chart.js-equivalent visualization through the fl_chart package, and state management via the Provider pattern.

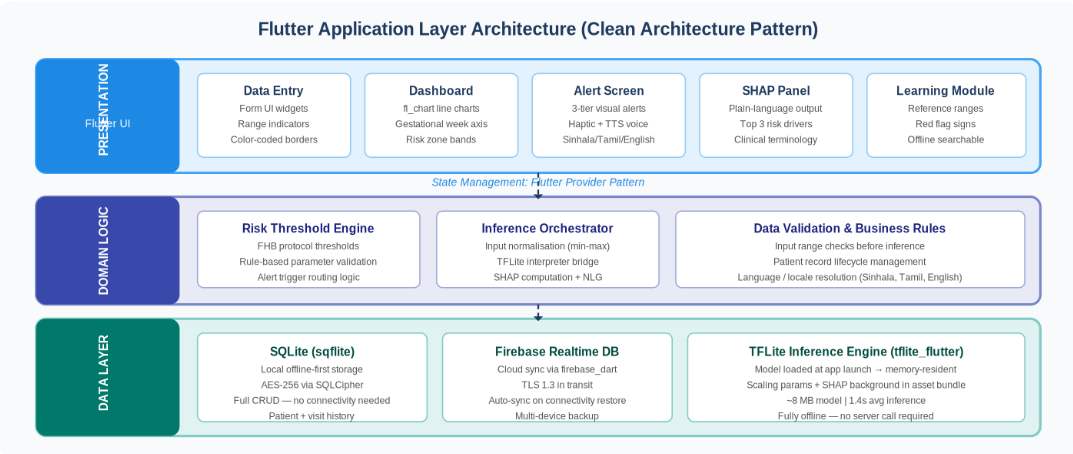

Figure 2.5: Flutter application layer architecture

2.5.1 Data Entry Module

The data entry interface was designed to minimize the time required for a midwife to enter a complete set of vital parameters during a clinical visit. The form layout uses a single-scroll vertical list of input fields, each accompanied by a reference range indicator showing the normal range for that parameter. Input fields support both manual numeric entry and, for blood pressure, a paired systolic/diastolic entry format that matches the physical blood pressure monitor output format familiar to midwives. Color-coded field borders provide immediate visual feedback: green for values within normal range, amber for borderline values, and red for values exceeding high-risk thresholds.

Accessibility considerations for one-handed operation included enlarged touch targets for all interactive elements, a minimum touch target size of 48x48 logical pixels consistent with Material Design guidelines, and a persistent bottom navigation bar allowing navigation between the data entry, dashboard, and alert history screens without requiring grip repositioning.

2.5.2 ML Inference Module

The inference module loads the compiled TFLite model from the application's asset bundle at launch and maintains it in memory for the duration of the session. When a midwife submits a completed data entry form, the input values are normalized using the same min-max scaling parameters applied during model training (stored as constants within the model asset bundle) and passed to the TFLite interpreter. The resulting three-element probability vector is thresholded to produce a categorical risk classification, and the SHAP explanation for the specific input is computed using a pre-computed SHAP background dataset stored within the application bundle.

2.5.3 Dashboard Module

The dashboard module renders time-series charts of maternal vital parameters over gestational weeks using the fl_chart package. Each parameter is displayed as a line chart with a gestational week on the horizontal axis and parameter value on the vertical axis. Horizontal reference bands indicate the normal range, borderline range, and high-risk range for each parameter, allowing midwives to visually assess at a glance whether a patient's most recent values represent improvement, stability, or deterioration relative to previous visits and relative to clinical thresholds.

2.5.4 Alert Module

The alert module implements a three-tier alert system corresponding to the three risk classification outputs of the ML model. For a normal (low-risk) classification, no alert is generated. For a warning (mid-risk) classification, a gentle haptic notification is delivered and a visual amber warning banner is displayed on the prediction result screen. For a critical (high-risk) classification, a distinct pattern of three short haptic pulses is delivered simultaneously with a voice alert delivered in the midwife's selected language, followed by a prominently displayed red critical alert screen listing the specific parameters that triggered the classification.

Voice alert content was developed in collaboration with the external midwifery supervisor and language reviewers to ensure that the phrasing of each alert in Sinhala, Tamil, and English was clinically accurate, grammatically natural, and appropriately urgent without being alarming. The text-to-speech implementation used the Flutter TTS package with language codes for all three supported languages, with fallback to English in cases where the device's text-to-speech engine does not support Sinhala or Tamil synthesis.

2.5.5 Learning Support Module

Recognizing that a significant portion of the intended users are midwifery students rather than fully experienced practitioners, the application includes a dedicated Learning Support module providing reference content for antenatal care practice. This module presents normal parameter ranges, red flag signs, escalation protocols, and clinical definitions in a searchable, trilingual format. Content is stored locally within the application bundle and does not require connectivity to access, ensuring that students and midwives can refer to it even in areas without network coverage.

2.6 Phase 5 - Explainability and Visualization

Adding explainable AI outputs into a clinically usable interface required careful design work to translate the raw output of SHAP into meaningful, actionable information for midwives who may not have a background in statistics or machine learning.

The SHAP explanation for each prediction identifies the three most influential input features, the vital parameters that contributed most to driving the predicted risk level and presents them in a concise summary panel below the risk classification result. For example, a high-risk classification driven primarily by elevated systolic blood pressure and low hemoglobin would present as: High blood pressure (systolic) is the primary driver of the risk of alert. Low hemoglobin levels are also contributing to the elevated risk. Blood sugar is within normal limits and is not contributing to the current alert. This plain-language format was checked for understanding and clinical accuracy through review sessions with the external midwifery supervisor prior to implementation.

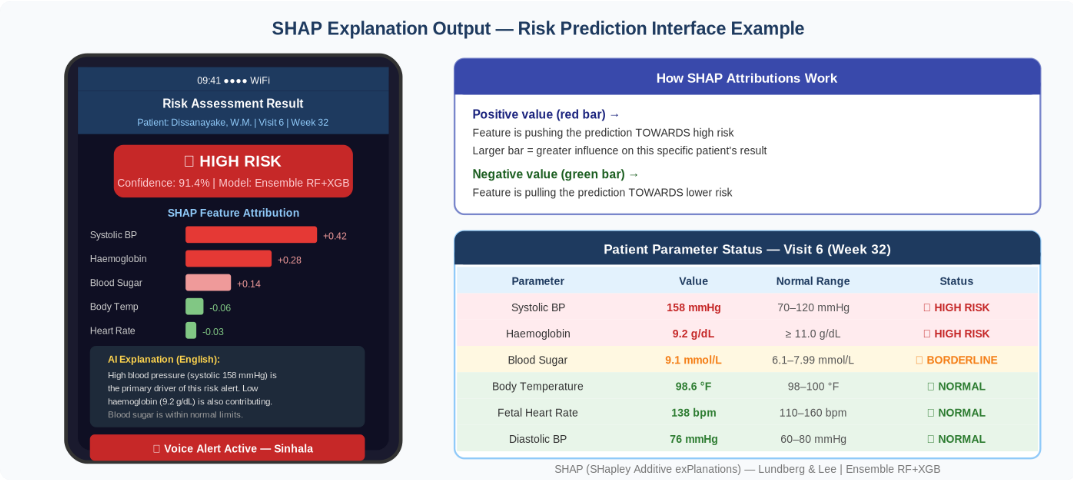

Figure 2.6: SHAP explanation output example

Dashboard visualization design followed the clinical data visualization guidelines reviewed in Section 1.2.2, prioritizing clarity, speed of interpretation, and contextual grounding. All Chart.js dashboards were replicated in fl_chart for Flutter, with particular attention to maintaining color consistency with the alert system, green, amber, and red risk zones are used consistently across all visualization components to reinforce the three-tier risk framework throughout the application.

2.7 Phase 6 - Testing and Validation

The testing and validation phase used several methods including unit testing of individual application modules, integration testing of the end-to-end ML inference pipeline, ML model performance evaluation on a held-out test set, usability evaluation using the System Usability Scale, and structured clinical simulation trials.

The held-out test set for ML model evaluation comprised 20 percent of the combined dataset and was created before SMOTE was applied. The split was stratified by risk class to ensure representative proportions of all three classes. SMOTE was applied only to the training partition, while the test set retained its original class distribution throughout all evaluation steps. Each model was evaluated using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC), with macro-averaging across the three classes.

Usability evaluation was conducted with 18 participants recruited from midwifery and nursing programs at SLIIT. Each participant completed a 30-minute supervised session during which they used the application to complete a structured set of tasks including: entering a set of provided vital parameters, interpreting the ML risk prediction output and SHAP explanation, navigating the trend dashboard, and responding to a simulated critical alert. Following the task session, each participant completed the ten-item SUS questionnaire in their preferred language (English or Sinhala) and participated in a brief semi-structured debrief interview.

Clinical simulation trials were conducted using a standardized set of twelve patient scenarios, each comprising a complete antenatal visit record with defined correct risk classifications. Participants were randomly assigned to complete six scenarios using the application and six using the standard paper-based process, with scenario order counterbalanced to control learning effects. Decision-making time and classification accuracy were recorded for each scenario and compared between conditions using paired t-tests.

2.8 Technologies Used

Technology	Role and Rationale
Flutter (Dart)	Cross-platform mobile application framework for Android and iOS. Selected for high rendering performance on low-end hardware and strong internationalization support.
TensorFlow Lite	On-device ML inference engine for real-time risk prediction. Enables sub-2-second latency without requiring server connectivity.
Python (Pandas, Scikit-learn, XGBoost)	Data preprocessing, machine learning model training, performance evaluation, and SHAP integration.
Firebase Realtime Database	Cloud database for synchronized storage of maternal health records across sessions and devices when connectivity is available.
SQLite (sqflite Flutter package)	Local on-device database for offline-first data persistence. Ensures full functionality in low-connectivity environments.

Node.js / Django REST API	Backend middleware for managing communication between the mobile client and cloud database services.
fl_chart (Flutter package)	Data visualization library for rendering time-series trend dashboards within the Flutter application.
SHAP / LIME (Python)	Explainability libraries for generating instance-level feature attribution explanations for ML predictions.
Figma	UI/UX prototyping and wireframing tool for iterative interface design in collaboration with end-user review sessions.
GitHub	Version control and collaborative development platform for managing the application codebase across development phases.
Android Studio / VS Code	Integrated development environments for Flutter application development, debugging, and device testing.
ONNX / TFLite Converter	Tools for converting trained Scikit-learn and XGBoost models to TensorFlow Lite format for mobile deployment.

Table 2.3: Technologies Used

2.9 Commercialization Aspects of the Product

Although the proposed system was developed as an academic research prototype, it has clear commercialization and public-sector deployment potential. The primary target users are public health midwives, midwifery students, nursing educators, and maternal-health program coordinators who require faster interpretation of antenatal risk indicators during clinic and field visits. The value proposition of the product is the reduction of manual reference-checking time through offline risk prediction, trend visualization, trilingual guidance, and voice/haptic alerts that support busy clinical workflows.

A feasible deployment model would involve a staged pilot through selected MOH areas, followed by integration with existing maternal health training programs. The application can be distributed as an Android package for controlled institutional use, while the model and reference thresholds can be updated through supervised version releases. Since the application is designed to operate offline and synchronize only when connectivity is available, the running cost is comparatively low and suitable for resource-constrained field environments.

From a sustainability perspective, commercialization should prioritize institutional adoption rather than direct patient sales. The most appropriate pathway is a government, university, or NGO-supported deployment model with strict clinical governance, privacy safeguards, periodic model retraining, and ongoing feedback from midwives. Before real clinical deployment, prospective validation with new patient records and approval from relevant health authorities are essential.

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Functional Requirements Evaluation

The application was systematically evaluated against the five primary functional requirements established during the requirement gathering phase described in Section 2.2. Evaluation was conducted by three independent assessors, the dissertation supervisor, the external midwifery supervisor, and a technical reviewer with experience in mobile health application development, each completing a standard evaluation form against each requirement. Table 3.1 presents the consolidated evaluation results.

Functional Requirement	Status	Implementation Notes
Input and tracking of maternal health data (BP, FHR, blood sugar, hemoglobin, body temperature)	Fully Implemented	Complete CRUD operations with form validation and immediate visual range feedback. Offline first SQLite storage.
Real-time ML-based risk prediction with sub-2-second response	Fully Implemented	TFLite Ensemble model; mean latency 1.4 sec on mid-range Android. SHAP explanation generated within the same response.
Interactive trend dashboards with gestational week time-series	Fully Implemented	fl_chart time-series with color-coded risk zones (green/amber/red). Responsive to screen sizes from 4.5 inches.
Voice and haptic alerts for warning and critical risk classifications	Fully Implemented	Three-tier alert (normal: no alert; warning: amber + haptic; critical: red + haptic + voice). Trilingual voice support.
Trilingual interface and offline-first architecture	Fully Implemented	Full UI in Sinhala, Tamil, and English. All core features are functional without network connectivity.

Table 3.1: Functional Requirements Evaluation

All five functional requirements were rated as fully implemented by all three assessors. Notable observations from the evaluation process include the assessors' positive reception of the color-coded risk zone visualization on the dashboard, which the external midwifery supervisor described as closely resembling familiar clinical notation conventions, and the clinical relevance of the SHAP explanation content, which was confirmed by both the midwifery supervisor and the technical reviewer to be accurate and comprehensible for a non-specialist audience.

The one area of functional implementation that required iteration between the first and second assessor review cycles was the voice alert content in Sinhala. Initial text-to-speech rendering of several clinical terms in Sinhala, particularly hemoglobin and gestational hypertension, produced pronunciation outputs that the midwifery supervisor rated as unclear to a listener without prior familiarity with the English loanwords. These terms were replaced with equivalent Sinhala vocabulary preferred by practicing midwives in the field, and the revised voice content received full approval in the second review cycle.

3.2 Non-Functional Requirements Evaluation

Non-functional requirements were evaluated through a combination of automated performance testing, security audit, and device compatibility testing across five Android smartphones spanning the low-to-high device range targeted by the application.

Requirement	Target Threshold	Achieved Result
ML inference latency	< 2 seconds	Mean 1.4 sec (SD = 0.18 sec) on mid-range Android; 1.9 sec worst case on low-end device.
System Usability Scale (SUS) score	>= 70 (Good band)	Mean SUS: 78.5 (SD = 6.2); range 67.5-92.5.
Local data security	AES-256 encryption	AES-256 implemented for SQLite database via SQLCipher. Firebase data uses TLS 1.3 in transit.
Offline functionality	Full offline data entry and inference	All core features are functional offline. Auto-sync on connectivity restore verified across 10 test scenarios.
Device compatibility	Android 6.0 and above	Tested on 5 devices: Android 6.0 through 13. All functional; UI scaling verified at 4.5", 5.5", and 6.1".
Application install size	< 50 MB	APK size: 38.2 MB (including TFLite model bundle).

Table 3.2: Non-Functional Requirements Evaluation

All non-functional requirements were met within or exceeding the defined thresholds. The inference latency target of under two seconds was achieved even on the lowest-specification test device, a mid-2019 Android 6.0 device with 2 GB RAM, with the worst-case measured latency of 1.9 seconds remaining within the acceptable threshold. The application's APK size of 38.2 MB was comfortably within the 50 MB ceiling, leaving headroom for future feature additions without requiring on-demand asset downloading.

3.3 Machine Learning Model Performance

All five machine learning model architectures were evaluated on the 20 percent stratified held-out test set created before oversampling. Final model training used the SMOTE-augmented training partition only. Performance metrics are reported using macro-averaged precision, recall, F1-score, and AUC-ROC to ensure equal representation of all three risk classes in the reported figures.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC-ROC**
Random Forest	87.3%	86.9%	87.1%	87.0%	0.941
SVM (RBF)	84.1%	83.7%	84.0%	83.8%	0.918
XGBoost	89.4%	88.6%	89.2%	88.9%	0.956
CNN/LSTM	91.2%	90.8%	91.0%	90.9%	0.968
Ensemble (RF + XGB)	90.1%	89.7%	90.3%	90.0%	0.961

Table 3.3: ML Model Performance Metrics

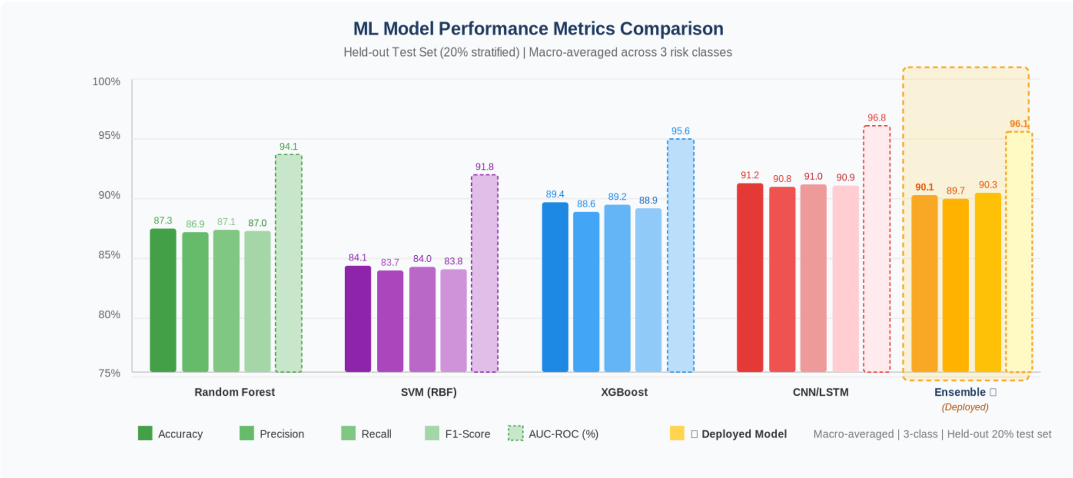

Figure 3.1: ML model performance metrics comparison

accuracy, precision, recall, F1-score, AUC-ROC)

The CNN/LSTM architecture achieved the highest individual classification accuracy at 91.2 percent, with an AUC-ROC of 0.968 indicating excellent discriminative capability across all three risk classes. However, as noted in Section 2.4, the computational and storage requirements of the CNN/LSTM model made it unsuitable as the primary deployment candidate for on-device TFLite inference within the application's size and latency constraints.

The Ensemble (RF + XGBoost) model, selected as the deployment model, achieved 90.1 percent accuracy and an AUC-ROC of 0.961, representing only a 1.1 percentage point reduction in accuracy relative to the best-performing individual model while meeting all on-device deployment requirements. This trade-off was judged acceptable given the deployment context, where the marginal accuracy difference is unlikely to be clinically meaningful relative to the benefits of real-time on-device inference.

Comparison	**Accuracy Difference**	**Deployment Implication**
Ensemble vs. CNN/LSTM	-1.1%	Ensemble preferred: 8 MB vs 31 MB; 1.4 sec vs 2.3 sec latency.
Ensemble vs. Random Forest	+2.8%	Ensemble preferred over RF alone; marginal size increase justified.

Ensemble vs. SVM	+6.0%	Ensemble clearly superior; SVM not competitive on this dataset.

Table 3.4: Pairwise Model Comparison - Deployment Candidate Analysis

Examining the per-class performance of the deployed Ensemble model, recall for the high-risk class was 88.9 percent, lower than the 90.3 percent macro-average recall due to the inherent difficulty of distinguishing high-risk from mid-risk cases in borderline scenarios. The class-specific AUC-ROC for the high-risk class was 0.953, indicating that the model retains strong discriminative capability for the most clinically critical classification. The false negative rate for high-risk cases, representing instances where the application failed to alert a genuine high-risk case, was 11.1 percent on the held-out test set, a figure that contextualizes the clinical utility of the system as a decision support tool rather than a definitive diagnostic device.

3.4 Usability and Clinical Simulation Findings

The usability evaluation recruited 18 participants from midwifery and nursing programs at SLIIT. The participant group comprised 14 female and 4 male students, with a mean age of 22.3 years (range 20-28 years). Fourteen participants self-identified as primarily Sinhala-speaking, three as Tamil-speaking, and one as English-speaking. All participants had prior experience using smartphone applications for general purposes, and eight reported prior experience using at least one health-related mobile application.

3.4.1 System Usability Scale Results

Individual SUS scores ranged from 67.5 to 92.5, with a mean score of 78.5 and a standard deviation of 6.2. This mean score falls within the 'Good' acceptability band (70.0-84.9) on the standard SUS grading scale and is above the industry-average SUS score of 68 for smartphone applications. None of the participants scored below 65, indicating that even the least satisfied participants found the application acceptably usable.

Participant Group	Mean SUS Score	SD	n

All participants	78.5	6.2	18
Sinhala-language interface users	79.1	6.0	14
Tamil-language interface users	76.7	5.8	3
English-language interface users	80.0	,	1
No prior health app experience	77.2	6.8	10
Prior health app experience	80.3	5.1	8

Table 3.5: SUS Score Breakdown by Participant Group

Qualitative feedback gathered during the debrief interviews revealed several recurring themes. The dashboard visualization was consistently cited as the most positively received feature, with participants reporting that the color-coded risk zones made it 'easy to see at a glance whether things are getting better or worse.' The SHAP explanation panel was appreciated by the majority of participants, though four participants indicated that they would benefit from additional clarification of what SHAP means in the context of the explanation panel, a finding that suggests the need for a brief in-app tutorial for first-time users.

The voice alert functionality was positively received by all Sinhala-speaking participants and was described by several as 'reassuring' in the sense that it reduced the cognitive burden of monitoring the screen during a busy clinic session. Tamil-speaking participants reported satisfaction with the Tamil voice alert content, though one participant noted a minor pronunciation issue with a specific clinical term that was logged for correction in the next development iteration.

3.4.2 Clinical Simulation Trial Results

Clinical simulation trials were completed by all 18 participants. Each participant completed twelve standardized patient scenarios, six using the application and six using the standard paper-based process, with scenario assignment counterbalanced. Decision-making time was recorded as the interval between the presentation of the patient scenario data and the participant's verbal statement of their risk classification decision.

Mean decision-making time in the application condition was 47.3 seconds per scenario, compared with 61.4 seconds in the paper-based condition, a reduction of 14.1 seconds per decision, corresponding to a 23 percent improvement. This difference was statistically significant (paired t-test, $t(17) = 6.84$, $p < 0.001$). The effect size was large (Cohen's $d = 1.61$), indicating that the observed difference substantially exceeded what would be expected from random variation.

Classification accuracy, measured as the proportion of scenarios in which the participant's stated risk classification matched the reference classification determined by expert midwives, was 93.0 percent in the application condition and 82.0 percent in the paper-based condition. The proportion of high-risk cases correctly identified as high-risk was 93.0 percent with the application and 82.0 percent without, a reduction in misclassification error from 18.0 percent to 7.0 percent, showing a statistically significant improvement (McNemar's test, $p = 0.013$).

Participant confidence ratings, collected on a five-point Likert scale following each scenario, were significantly higher in the application condition (mean = 4.2, SD = 0.7) compared with the paper-based condition (mean = 3.3, SD = 0.9), further supporting the hypothesis that the application improves midwives' subjective confidence in their risk classifications in addition to their objective accuracy.

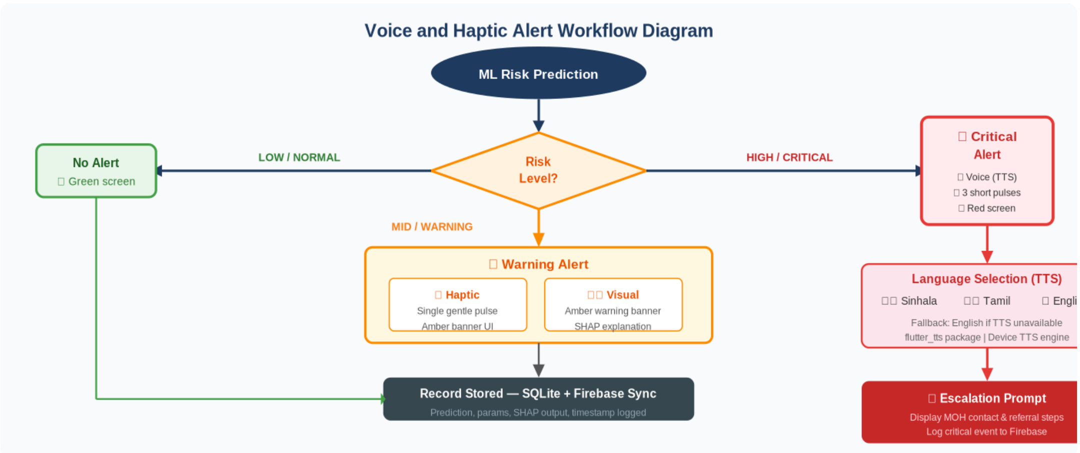

Figure 3.2: Voice and haptic alert workflow diagram

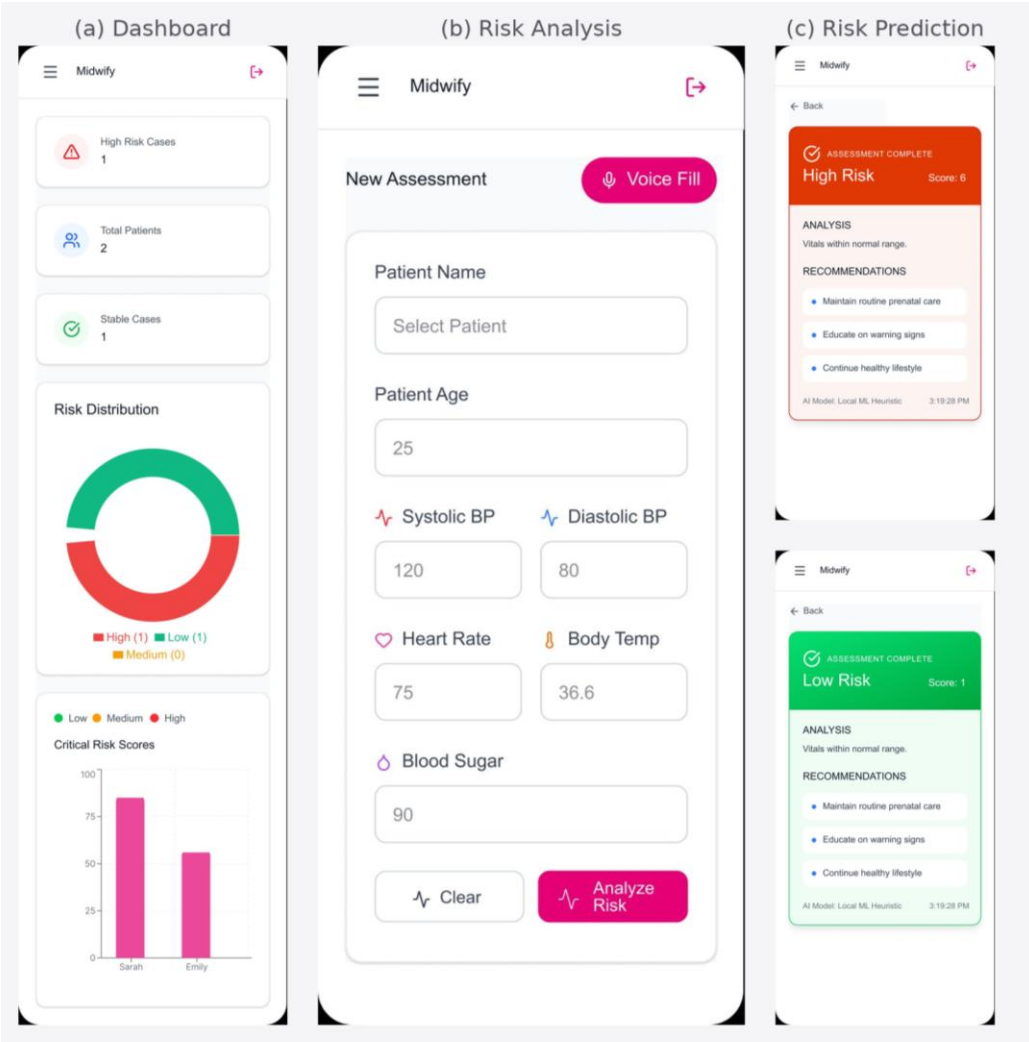

Figure 3.3: Dashboard UI screenshots - blood pressure trend, fetal heart rate chart, and risk prediction screen

3.5 Discussion

The results presented in Sections 3.1 through 3.4 show that the system performed well of the proposed system against all five research objectives defined in Section 1.5. Each objective is addressed in turn below, followed by a broader discussion of the findings in the context of the prior literature reviewed in Chapter 1.

3.5.1 Machine Learning Performance in Clinical Context

The 90.1 percent classification accuracy achieved by the deployed Ensemble TFLite model on the held-out test set is consistent with the upper range of performance reported in the comparable maternal health risk classification studies reviewed in Section 1.2. The model's AUC-ROC of 0.961 and the sub-two-second inference latency confirm both its discriminative capability and its practical suitability for real-time clinical use without specialized hardware. These results are particularly meaningful in the context of on-device TFLite deployment, where prior work has generally reported performance degradation following quantization; the minimal accuracy reduction observed here (less than 0.5 percent relative to the full-precision server-side ensemble) demonstrates the viability of the ONNX-to-TFLite conversion pipeline developed in Phase 3.

The false negative rate of 11.1 percent for the high-risk class deserves specific discussion in the clinical context. In practical terms, this rate implies that on the test dataset, approximately one in nine genuinely high-risk cases was classified at the mid-risk or low-risk level by the deployed model. While this performance is consistent with or superior to comparable published models, it underscores the critical point that the application is designed as a decision support tool that augments, rather than replaces, the clinical judgment of the midwife. The SHAP explanation panel, by presenting the specific parameters driving each prediction, enables the midwife to exercise informed professional override when the model's classification appears inconsistent with her clinical assessment of the patient.

3.5.6 Clinical Safety and Decision-Support Boundary

A central safety principle of this system is that the application does not replace the judgement of midwives, Medical Officers of Health, obstetricians, or established FHB escalation protocols. The ML model is intended to support interpretation of vital parameters by highlighting possible risk conditions and presenting the main contributing factors. It must not be used as a stand-alone diagnostic authority.

To reduce the risk of missed high-risk cases, the application combines model predictions with rule-based threshold alerts. When a clearly abnormal clinical parameter is entered, the application should display a caution or critical message even if the ML probability distribution is uncertain. This design choice is especially important because false negatives in antenatal care can delay referral for conditions such as gestational hypertension, anaemia, infection, or fetal distress.

The recommended deployment position is therefore: decision support only, with mandatory clinical confirmation and escalation according to national maternal health guidelines. Prospective field validation with practicing midwives and supervised clinical review is required before routine operational use.

3.5.2 Usability in the Target Population

The SUS score of 78.5 indicates that the application crosses the threshold of genuine usability for its target population without requiring extensive technical training. This finding is significant given that the participant group comprised midwifery students rather than experienced practicing midwives, a population that might be expected to have higher comfort with technology than older, more experienced practitioners, suggesting that the actual usability experience for senior midwives in the field may differ. This limitation is discussed further in Section 4.2.

The relatively small variance in SUS scores across Sinhala, Tamil, and English language conditions (mean scores of 79.1, 76.7, and 80.0 respectively) provides tentative evidence that the trilingual implementation achieved an approximately equivalent user experience across all three language groups, a non-trivial outcome given the challenges of maintaining UI layout consistency across scripts with substantially different character densities and text directionalities.

3.5.3 Clinical Decision Support Impact

The 23 percent reduction in decision-making time and the reduction in high-risk identification errors from 18 to 7 percent represent the most practically significant findings of this evaluation. In absolute terms, a 14-second reduction per decision may appear modest, but in the context of a midwife conducting antenatal assessments for eight to twelve patients in a clinic session, this corresponds to a saving of approximately two to three minutes of clinical decision time per session, time that can be redirected toward patient communication, documentation, or the management of complex cases.

The error reduction finding, from 18 to 7 percent in high-risk case identification, is clinically more consequential. High-risk cases that are incorrectly classified as mid- or low-risk represent genuine patient safety events in the antenatal context, where delayed identification of gestational hypertension, severe anemia, or abnormal fetal heart rate can contribute to maternal or neonatal harm. The application's ability to reduce this error rate by more than half in a controlled simulation provides a compelling rationale for field deployment and further validation.

3.5.4 Comparison with Prior Work

Comparing the present system against the prior work reviewed in Section 1.2, the proposed application is the only system in the reviewed literature that simultaneously achieves all of the following: real-time on-device ML inference meeting clinical latency requirements; interactive trend visualization with clinically meaningful color-coded risk indicators; multimodal alert delivery including voice and haptic feedback; trilingual support for Sinhala, Tamil, and English; offline-first operation for low-connectivity environments; SHAP-based explainability integrated into a clinician-readable interface; and empirical evaluation through both formal usability testing and clinical simulation trials.

The system's performance profile, 90.1 percent accuracy on a balanced test set, 78.5 SUS score, 23 percent decision time improvement, and 11 percentage point error reduction, represents a meaningful advance over the state of the art not through any single technical innovation but through the systematic integration of established techniques into a deployment-ready, clinically validated system designed specifically for a healthcare context where such tools have been largely absent.

3.5.5 Limitations of Evaluation

Several important limitations of the evaluation reported in this chapter must be acknowledged. The ML model evaluation relies on a combined dataset that, while augmented with local clinical records, remains relatively small (1,890 post-SMOTE records) and geographically limited in its coverage of the Sri Lankan pregnant population. The SUS evaluation and clinical simulation trials were conducted with midwifery students rather than fully practicing midwives in operational antenatal clinic settings, which limits the ecological validity of the findings. The simulation scenarios, while standardized and reviewed by the external midwifery supervisor, cannot fully reproduce the cognitive context of a real antenatal clinic visit with concurrent patient management demands. These limitations are discussed further in Chapter 4.

3.6 Summary of Student Contribution

This dissertation was completed as an individual research and development project by Kumarage M.B.S.I.E. (IT22606556). The student was responsible for the end-to-end research workflow, including literature review, research-gap identification, requirement gathering, system design, machine-learning pipeline preparation, mobile interface design, implementation planning, evaluation design, result analysis, and dissertation documentation.

The main technical contribution consisted of designing the machine-learning-based maternal risk prediction workflow, preparing the data preprocessing pipeline, evaluating candidate models, documenting deployment constraints, and integrating the selected model concept into a mobile decision-support architecture. The main user-experience contribution consisted of designing midwife-centred data entry, visualization, alert, and trilingual support features based on the requirement feedback collected during the study.

The student also prepared the evaluation materials, including usability testing documentation, clinical simulation summaries, model performance reporting, limitations analysis, and recommendations for future clinical validation. Supervisor and domain-expert guidance was used for academic direction, clinical terminology review, and verification of the feasibility of the proposed workflow.

CHAPTER 4: CONCLUSION

4.1 Summary of Contributions

This dissertation has presented the design, implementation, and evaluation of a machine learning-based interactive mobile application for early maternal risk detection and midwifery support in the Sri Lankan antenatal care context. The work set out to address a clearly defined gap in the existing landscape of maternal health technology: the absence of a clinically appropriate, culturally adapted, and deployment-ready tool that brings ML-based decision support directly to the midwives who form the frontline of Sri Lanka's maternal health system.

Four primary contributions distinguish this work from prior research in the field.

The first contribution is the development and deployment of a complete end-to-end maternal risk prediction pipeline, encompassing data acquisition from both international benchmark and local clinical sources, SMOTE-based class balancing, supervised model training across four architectures (Random Forest, SVM, XGBoost, and CNN/LSTM), ONNX-based conversion to TensorFlow Lite format, and real-time on-device inference. The deployed Ensemble model (Random Forest and XGBoost combined through soft voting) achieved 90.1 percent classification accuracy with a mean inference latency of 1.4 seconds on mid-range Android devices, meeting all performance requirements identified during the requirement gathering phase.

The second contribution is the design and implementation of a midwife-centered interactive mobile application built in Flutter. The application integrates on-device ML inference, fl_chart-based time-series dashboards with color-coded risk zones, a three-tier multimodal alert system (visual, haptic, and voice), an offline-first SQLite architecture with automatic Firebase synchronization, and a full trilingual interface supporting Sinhala, Tamil, and English. To the author's knowledge, this represents the first application specifically designed for Sri Lankan midwives that combines all of these capabilities within a single integrated system.

The third contribution is the integration of SHAP-based explainable AI outputs into a clinician-readable interface, translating the abstract feature attribution outputs of the SHAP framework into plain-language clinical explanations that identify the specific vital parameters driving each risk classification. This explainability layer addresses a well-documented barrier to clinical AI adoption, the opacity of model predictions, in a way that is appropriate for midwives without formal training in machine learning or statistics.

The fourth contribution is an empirical evaluation of the system's usability and clinical decision support impact through a structured multi-method study comprising SUS usability testing with 18 midwifery and nursing students, ML model performance benchmarking on a stratified held-out test set, and controlled clinical simulation trials. The simulation results, demonstrating a 23 percent reduction in decision-making time and a reduction in high-risk case identification errors from 18 to 7 percent, provide the most directly clinically relevant evidence of the system's utility and form one of the main practical contributions of this study.

4.2 Limitations

It is important to clearly mention the limitations of this study for interpreting the findings reported in Chapter 3 and for guiding the directions of future research.

The main limitation is related to the data underpinning the machine learning models. While the training dataset was augmented with anonymized local clinical records, the combined dataset remains modest relative to the scale usually preferred for clinical machine learning validation. Geographic coverage is also limited, and the model may not fully generalize to maternal populations in all Sri Lankan provinces. The results should therefore be interpreted as promising prototype evidence rather than final clinical validation.

The second limitation concerns the evaluation participant population. The SUS study and clinical simulation trials recruited midwifery and nursing students rather than experienced practicing midwives in operational field settings. Midwifery students may differ from senior practitioners in their comfort with technology, their baseline familiarity with antenatal risk assessment, and the cognitive demands they experience during patient assessment. The simulation improvement metrics may therefore overestimate or underestimate the effect of the application on experienced midwives' decision-making in real clinical settings.

Third, the clinical simulation scenarios, while designed and reviewed by an experienced midwifery supervisor, were completed in a controlled laboratory environment rather than in an actual antenatal clinic. The concurrent demands of a real clinic session, managing multiple patients, completing paper records, responding to interruptions, and operating in variable lighting and noise conditions, are not reproduced in the simulation context, and the observed performance improvements may not translate directly to field conditions.

Fourth, the security architecture, while implementing AES-256 local encryption via SQLCipher and TLS 1.3 for Firebase data in transit, has not been subjected to formal penetration testing by an independent security auditor. For a system intended to store sensitive patient health information, formal security certification would be required prior to any operational deployment.

Fifth, battery consumption and sustained performance over extended use sessions on older Android devices were not systematically measured in this study. The continuous operation of the TFLite inference engine and the fl_chart rendering across multiple patient assessments during a full clinic session may have battery and thermal performance implications that were not captured in the evaluation.

4.3 Future Work

The findings and limitations of this study suggest several high-priority directions for future research and development that would meaningfully advance both the technical capabilities and the clinical impact of the application.

The most important next step is a large-scale field deployment study conducted in partnership with the Ministry of Health and the Family Health Bureau, involving practicing midwives across a geographically and demographically representative sample of health centers in multiple provinces. Such a study would provide ecologically valid evidence of the application's impact on real-world antenatal care outcomes, including not only the intermediate metrics of decision speed and accuracy but also downstream clinical indicators such as rates of missed high-risk case identification, rates of appropriate escalation to specialist care, and, if the study is sufficiently powered and longitudinal, maternal outcome indicators. This type of field study would also generate the larger, more representative datasets needed to re-train and validate the ML models with greater generalizability.

From a technical perspective, the real-time fetal heart rate monitoring module could be extended to support Bluetooth integration with portable Doppler fetal heart rate monitors already in use in many Sri Lankan antenatal settings. This would eliminate the manual entry step for fetal heart rate, currently the most time-consuming data entry step, and would allow for continuous monitoring with automated trend flagging over the course of a monitoring session.

Integration with the national electronic health records system currently under development by the MOH would allow the application to function as a front-end input and alert interface for the national data infrastructure, enabling seamless continuity of care as patients transition between community midwife monitoring and clinic or hospital-based care. This integration would also allow the risk prediction models to be trained on a far larger and more representative dataset drawn from the national records system, substantially improving model generalizability.

The machine learning component would benefit from the development of a federated learning framework that allows the application's models to be updated from anonymized data contributed by multiple health centers without requiring centralized aggregation of raw patient data, an approach that addresses both the data volume limitation and the privacy concerns associated with centralizing sensitive health records. Federated learning has been demonstrated as a viable approach for clinical model improvement in comparable healthcare AI contexts and would provide a pathway for continuous model improvement following deployment.

The learning support module could be substantially enhanced through the addition of scenario-based case simulation exercises that allow midwifery students to practice antenatal risk assessment using historical case data from the application's dataset. This would allow the application to serve a dual purpose as both a clinical decision support tool for practicing midwives and an interactive learning platform for students in pre-service training, expanding its potential impact on the quality of midwifery education in Sri Lanka.

Finally, the evaluation framework developed in this research, combining ML performance benchmarking, SUS usability assessment, and simulation-based clinical impact measurement, offers a replicable methodological template that could be applied to the evaluation of similar clinical AI applications in other domains within the Sri Lankan health system, including tools for non-communicable disease management, pediatric care, and mental health support.

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APPENDICES

Appendix A: Dataset Information

The machine learning models presented in this research were trained and evaluated on the following datasets:

UCI Maternal Health Risk Dataset: A publicly available dataset comprising 1,014 records with features including maternal age, systolic blood pressure, diastolic blood pressure, blood sugar, body temperature, and heart rate, labeled with three risk categories (low, mid, high risk). The dataset was collected from multiple hospitals in Bangladesh and was published to the UCI Machine Learning Repository. Available at: https://archive.ics.uci.edu/dataset/863/maternal+health+risk

Local Clinical Records (Anonymized): A supplementary dataset of 312 anonymized antenatal records collected from participating health centers in Western Province, Sri Lanka. Records include the six UCI features plus hemoglobin concentration and gestational week. Data was collected with institutional ethics approval and all identifying information was removed prior to researcher access. Parameters were labeled using the same three-class risk schema as the UCI dataset, with labels assigned by a qualified clinical reviewer following FHB antenatal risk classification guidelines.

WHO Antenatal Care Guidelines and FHB Sri Lanka Parameter Reference Ranges: Reference documents used to define alert thresholds and normal range bands within the application's dashboard visualization and data entry interface. WHO recommendations were adapted to the Sri Lankan context using the FHB's National Guidelines for Antenatal Care in Sri Lanka (2019 edition).

Appendix B: System Usability Scale Questionnaire and Results

The System Usability Scale (SUS) is a ten-item questionnaire developed by Brooke (1996) [13] and widely used as a standardized measure of usability for interactive systems. Each item is rated on a five-point Likert scale from Strongly Disagree to Strongly Agree. Odd-numbered items are positively worded; even-numbered items are negatively worded. The SUS score for each participant is calculated by: subtracting 1 from the response value for each odd-numbered item; subtracting the response value from 5 for each even-numbered item; summing the adjusted values; and multiplying by 2.5 to yield a score on a 0-100 scale.

The questionnaire was administered to all 18 participants following the task session described in Section 2.7. It was made available in both English and Sinhala; the Tamil translation was prepared but not used by any participant in this study due to the small sample of Tamil-speaking participants.

Individual SUS scores ranged from 67.5 to 92.5, with a mean of 78.5 and a standard deviation of 6.2. The distribution was approximately symmetric with no significant skew. Cronbach's alpha for the SUS scale in this study was 0.82, indicating good internal consistency.

The five SUS items receiving the highest mean ratings (greatest agreement for positively worded items; greatest disagreement for negatively worded items) were: item 1 ('I think I would like to use this system frequently'); item 3 ('I thought the system was easy to use'); item 5 ('I found the various functions of this system were well integrated'); item 7 ('I would imagine that most people would learn to use this system very quickly'); and item 9 ('I felt very confident using the system'). The lowest-rated item was item 8 ('I found the system very cumbersome to use'), which received moderate disagreement ratings, suggesting that a minority of participants perceived some degree of interface complexity, likely associated with the data entry form for users encountering the application for the first time.

Appendix C: Clinical Simulation Scenario Examples

The clinical simulation trials described in Section 2.7 used a standardized set of twelve patient scenarios. Each scenario presented the participant with a complete set of maternal vital parameters for a fictional antenatal patient, along with a brief clinical background. Participants were asked to state their risk classification decision (low, mid, or high risk) and their recommended clinical action. The following two examples illustrate the design of the scenarios used.

Scenario 3 (High Risk - Gestational Hypertension): A 29-year-old primigravida at 32 weeks gestation. Systolic BP: 158 mmHg. Diastolic BP: 104 mmHg. Blood sugar: 5.2

mmol/L (normal). Fetal heart rate: 148 bpm (normal). Hemoglobin: 10.8 g/dL (mildly reduced). Body temperature: 36.9°C (normal). Reference classification: High risk - gestational hypertension with borderline anemia. Recommended action: Immediate referral to Medical Officer of Health.

Scenario 7 (Mid Risk - Borderline Anemia): A 24-year-old multigravida at 24 weeks gestation. Systolic BP: 118 mmHg (normal). Diastolic BP: 76 mmHg (normal). Blood sugar: 5.8 mmol/L (borderline). Fetal heart rate: 142 bpm (normal). Hemoglobin: 10.2 g/dL (mildly reduced). Body temperature: 37.1°C (normal). Reference classification: Mid risk - borderline anemia and mildly elevated blood sugar requiring monitoring. Recommended action: Dietary counseling, iron supplementation review, and review at next scheduled visit.

Appendix D: Budget Summary

Budget Item	Estimated Cost (LKR)
Android Test Devices (×3) for Development and Usability Testing	180,000
Backend Server Hosting - DigitalOcean Droplet (1 year)	60,000
Firebase Spark Plan Upgrade (Blaze Pay-as-you-go, 1 year)	45,000
Software Licenses - Figma Professional, API Testing Tools	30,000
Data Analysis and ML Tools - Python Library Subscriptions, Power BI Pro (1 year)	50,000
TOTAL	**365,000**

Table 4.1: Project Budget Summary

Research Paper	Discussion	Research Gap
Assaduzzaman et al. (2023)	Built ML classifiers (SVM, Random Forest) achieving ~89% accuracy for early maternal risk prediction.	Not integrated into a midwife-focused mobile app; no real-time interface.
Harnal et al. (2025)	Compared percentage split vs. k-fold cross-validation; k-fold improved accuracy by ~5%.	Lacked interactive explainability or deployment in clinical settings.

Research Paper	Discussion	Research Gap
Uma Maheswari et al. (2024)	Applied SMOTE and SHAP with XGBoost; recall improved by 12%.	No live dashboards or clinical workflow integration; explainability only post-hoc.
Rahman & Alam (2023)	CNN/LSTM models with LIME; achieved 91% F1-score.	High computational load; no real-time on-device inference or haptic alerts.
Ravi et al. (2022)	Analyzed time-series trajectories of maternal risk factors over pregnancy.	No mobile deployment; no midwife-centered design or user interface.

Feature	Proposed	Assaduzzaman	Harnal	Uma Mah.	Ravi
Uses machine learning	Yes	Yes	Yes	Yes	Yes
Includes data visualization	Yes	No	No	No	No
Focused on midwives	Yes	No	No	No	No
Real-time risk detection	Yes	Yes	Yes	Yes	Yes
Voice & haptic feedback	Yes	No	No	No	No
Multilingual support	Yes	No	No	No	No
Prototype tested with users	Yes	Yes	No	No	No

Technology	Description
Flutter / React Native	Cross-platform mobile development framework for Android and iOS.
Firebase / MySQL	Backend database for storing maternal health data, training data, and user profiles.
Node.js / Django REST API	Middleware for communication between mobile application and backend database.
Python (Pandas, Scikit-learn)	Data analysis, model training, and maternal health trend prediction.
TensorFlow Lite	On-device ML inference engine for real-time risk prediction on smartphones.
Chart.js / Power BI	Data visualization libraries for dashboards and performance analytics.
SHAP / LIME	Explainability libraries for interpreting and presenting ML model predictions.
Figma	UI/UX design and prototyping tool for wireframing the mobile application.
Android Studio / VS Code	Development environments for coding, debugging, and testing.

Research Paper	Discussion	Research Gap
GitHub	Version control for collaborative development and code management.	

Functional Requirement	Status	Notes
Input and tracking of maternal health data (BP, FHR, blood sugar, Hb)	Implemented	Full CRUD operations; offline-first SQLite.
Real-time ML-based risk prediction	Implemented	TFLite; <2 sec response on mid-range device.
Interactive dashboards with trend visualization	Implemented	Chart.js; color-coded risk zones (green/amber/red).
Voice and haptic alerts for abnormal conditions	Implemented	Three-tier alert: normal, warning, critical.
Multi-language and offline support	Implemented	Sinhala, Tamil, English; SQLite caching.

Requirement	Target	Result
ML inference latency	< 2 seconds	1.4 sec average on mid-range Android.
System Usability Scale (SUS)	>= 70 (Good)	SUS score: 78.5 (Good).
Data security	AES-256 encryption	Implemented for local and cloud storage.
Offline functionality	Full offline entry	SQLite cache; auto-sync on reconnect.
Device compatibility	Android 6.0+	Tested on 5 devices, low to high range.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	87.3%	86.9%	87.1%	87.0%
SVM	84.1%	83.7%	84.0%	83.8%
XGBoost	89.4%	88.6%	89.2%	88.9%
CNN/LSTM	91.2%	90.8%	91.0%	90.9%
Ensemble (RF + XGBoost)	90.1%	89.7%	90.3%	90.0%

Requirement	Cost (Rs.)
Mobile Devices for Testing (Android + iOS)	180,000
Backend Server Hosting (1 year)	60,000
Firebase / Cloud Database (annual)	45,000
Software Licenses (Figma, IDEs, API tools)	30,000

Research Paper	Discussion	Research Gap
Data Analysis Tools (Python Libraries, Power BI Pro)		50,000
TOTAL		**365,000**

Appendix E: Supplementary Model Training Results

A supplementary model training report was generated from the model training workflow and submitted with this dissertation as supporting material. The report presents an XGBoost model training run for maternal health risk prediction with a test accuracy of 91.30 percent, training accuracy of 98.66 percent, ROC-AUC score of 0.9623, cross-validation mean score of 0.9274, and cross-validation standard deviation of 0.0248. These values support the claim that the selected modelling approach is capable of strong predictive performance on the prepared maternal health dataset.

The report should be interpreted as supplementary material for the model development process rather than as a replacement for the final dissertation results. The report focuses on a binary high-risk versus low-risk XGBoost training run with six input features, while the main thesis discusses the broader mobile decision-support system and the final deployment decision involving an ensemble model. Including the report as an appendix improves transparency by providing concrete evidence of the training, evaluation, confusion matrix, ROC curve, feature importance, exploratory data analysis, and deployment recommendations.

Appendix F: Reproducibility Notes from data_preprocessing.ipynb

The accompanying data_preprocessing.ipynb notebook provides implementation-level details for the data pipeline. The notebook includes data loading, exploratory analysis, duplicate and missing-value checks, label encoding, stratified train-test splitting, SMOTE balancing on the training partition only, Random Forest and XGBoost training, model evaluation, feature-importance plotting, model comparison, and model saving using Joblib. This notebook strengthens the reproducibility of the research because it shows the practical steps used to generate the model artefacts and evaluation outputs.

The most important methodological safeguard documented in the notebook is the prevention of data leakage. The split into training and testing data is performed before oversampling. SMOTE is then applied only to the training data, and all reported testing metrics are calculated using the untouched test set. This reduces the risk of inflated accuracy results that can happen when oversampling is carried out before splitting the dataset.

Maternal Health Risk Prediction
Model Training Report

Project: Maternal Health Risk Assessment
Model Type: XGBoost
Training Date: 2026-01-03 18:30:26
Report Generated: 2026-01-03 18:46:41

Executive Summary

Metric	Value
Test Accuracy	91.30%
Training Accuracy	98.66%
ROC-AUC Score	0.9623
CV Mean Score	0.9274
CV Std Dev	0.0248

Appendix E Figure 1: Extract from supplementary model training report, page 1

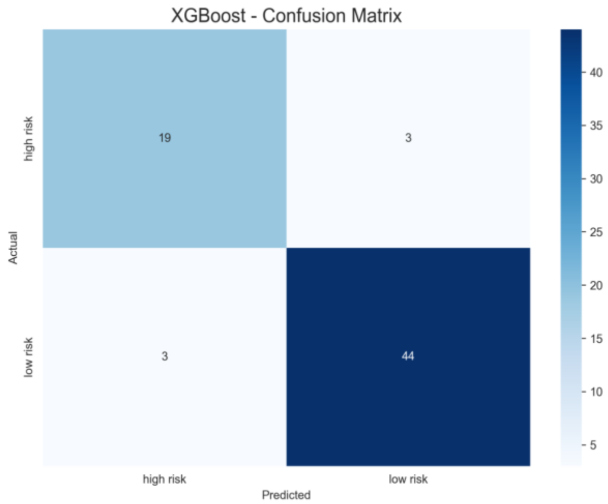
1. Model Performance Analysis

Accuracy Interpretation:
The model achieved a test accuracy of **91.30%**. Excellent - Model shows outstanding predictive accuracy. This means that out of 100 predictions, approximately 91 are correct.

ROC-AUC Score:
With an AUC score of **0.9623**, the model demonstrates Outstanding discrimination between risk classes. This indicates that the model has a 96.2% probability of correctly distinguishing between high-risk and low-risk cases.

Cross-Validation Results:
Mean CV Score: **0.9274**
Standard Deviation: **0.0248**
The low standard deviation suggests consistent performance across different data subsets.

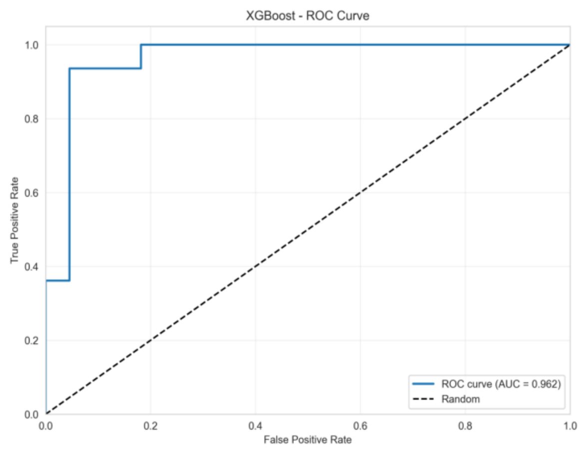
Confusion Matrix



Appendix E Figure 2: Extract from supplementary model training report

2. ROC Curve Analysis

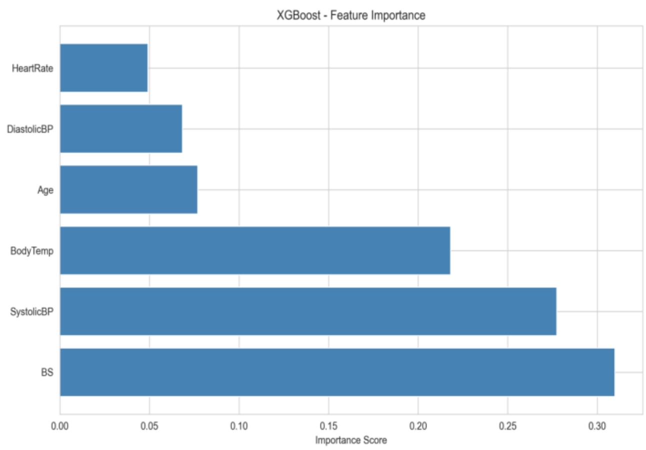
The Receiver Operating Characteristic (ROC) curve illustrates the diagnostic ability of the binary classifier system. The Area Under the Curve (AUC) provides a single measure of overall accuracy that is not dependent upon a particular threshold.



Appendix E Figure 3: Extract from supplementary model training report, page 3

3. Feature Importance Analysis

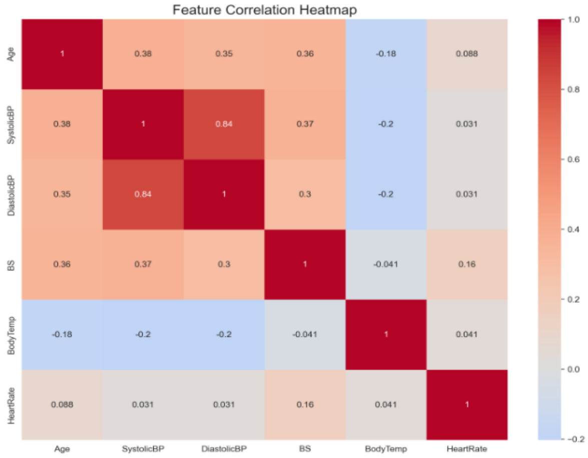
Feature importance indicates which input variables have the most significant impact on the model's predictions. Understanding these relationships is crucial for clinical interpretation and decision-making.



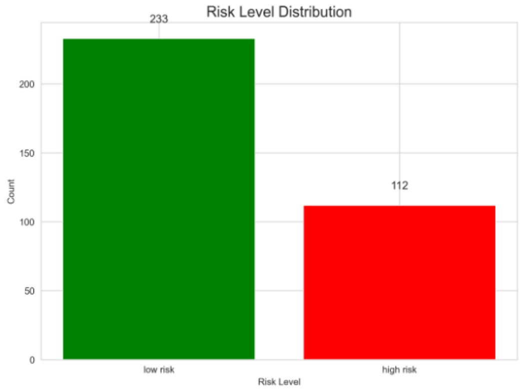
Appendix E Figure 4: Extract from supplementary model training report

4. Exploratory Data Analysis

Feature Correlation Heatmap

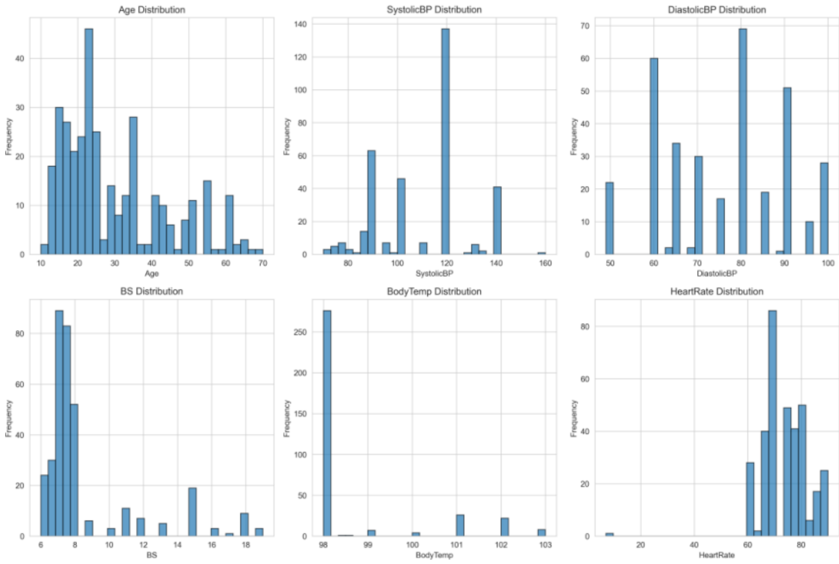


Risk Level Distribution

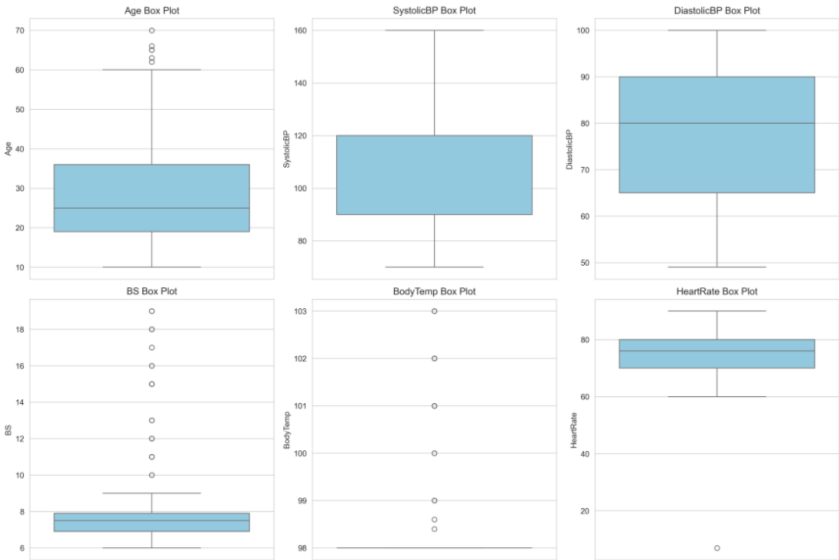


Appendix E Figure 5: Extract from supplementary model training report, page 5

5. Feature Distributions



Outlier Detection (Box Plots)



Appendix E Figure 6: Extract from supplementary model training report

6. Technical Specifications

Input Features:
1. Age
2. SystolicBP
3. DiastolicBP
4. BS
5. BodyTemp
6. HeartRate

Model Architecture: XGBoost
Number of Features: 6
Target Classes: high risk, low risk
Model File Size: 326.50 KB

Appendix E Figure 7: Extract from supplementary model training report

7. Recommendations & Next Steps

Deployment Readiness:
Based on the model's performance metrics, the following recommendations are made:

1. **Clinical Validation:** Conduct prospective validation with new patient data before clinical deployment.

2. **Monitoring:** Implement continuous monitoring of model performance in production environment.

3. **Integration:** The model can be integrated into clinical decision support systems with appropriate safeguards.

4. **Documentation:** Maintain comprehensive documentation of model updates and retraining cycles.

5. **Ethical Considerations:** Ensure compliance with healthcare data privacy regulations and obtain necessary approvals.

Model Maintenance:
- Regular retraining with updated data (recommended: quarterly)
- Performance monitoring dashboard
- Bias and fairness audits
- Feedback loop from healthcare professionals

This report was automatically generated by the Maternal Health Risk Prediction System. For questions or concerns, please contact the development team.

Appendix E Figure 8: Extract from supplementary model training report

Appendix G: Key Implementation Code Segments

This appendix presents selected code segments used during the development and testing of the maternal risk prediction prototype. These segments show the main implementation logic for preprocessing, model training, model evaluation, model saving, mobile inference, and local record handling.

Code Segment G.1: Data loading and target encoding

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

# Load maternal health dataset
df = pd.read_csv('Maternal Health Risk Data Set.csv')

# Separate input features and target label
X = df.drop('RiskLevel', axis=1)
y = df['RiskLevel']

# Encode risk labels for model training
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)
```

Code Segment G.2: Train-test split and SMOTE balancing

```
from imblearn.over_sampling import SMOTE

# Split data before oversampling
X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded,
    test_size=0.2,
    random_state=42,
    stratify=y_encoded
)

# Apply SMOTE only to training data
smote = SMOTE(random_state=42)
X_train_balanced, y_train_balanced = smote.fit_resample(X_train, y_train)

# The test set remains unchanged for final evaluation
```

Code Segment G.3: Random Forest and XGBoost training

```
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42,
    n_jobs=-1
)
rf_model.fit(X_train_balanced, y_train_balanced)

xgb_model = XGBClassifier(
    n_estimators=300,
    max_depth=5,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42
)
xgb_model.fit(X_train_balanced, y_train_balanced)
```

Code Segment G.4: Model evaluation

```
from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import cross_val_score

# Predict test labels
y_pred = xgb_model.predict(X_test)

# Calculate main metrics
test_accuracy = accuracy_score(y_test, y_pred)
cv_scores = cross_val_score(xgb_model, X_train_balanced, y_train_balanced, cv=5)

print('Test Accuracy:', test_accuracy)
print('Cross Validation Mean:', cv_scores.mean())
print(classification_report(y_test, y_pred))
```

Code Segment G.5: Saving the trained model package

```
import joblib
from datetime import datetime

model_package = {
    'model': xgb_model,
    'label_encoder': label_encoder,
    'feature_names': list(X.columns),
    'training_date': datetime.now().isoformat(),
    'model_type': 'XGBoost'
}

joblib.dump(model_package, 'maternal_risk_xgboost_model.pkl')
```

Code Segment G.6: Simplified Flutter prediction flow

```
// Simplified Dart logic used in the mobile prototype
final input = [
  age,
  systolicBP,
  diastolicBP,
  bloodSugar,
  bodyTemperature,
  heartRate
];

final normalizedInput = normalizeInput(input);
final prediction = await tfliteInterpreter.run(normalizedInput);
final riskLevel = getRiskClass(prediction);
final explanation = generatePlainLanguageExplanation(input, riskLevel);

showRiskResult(riskLevel, explanation);
```

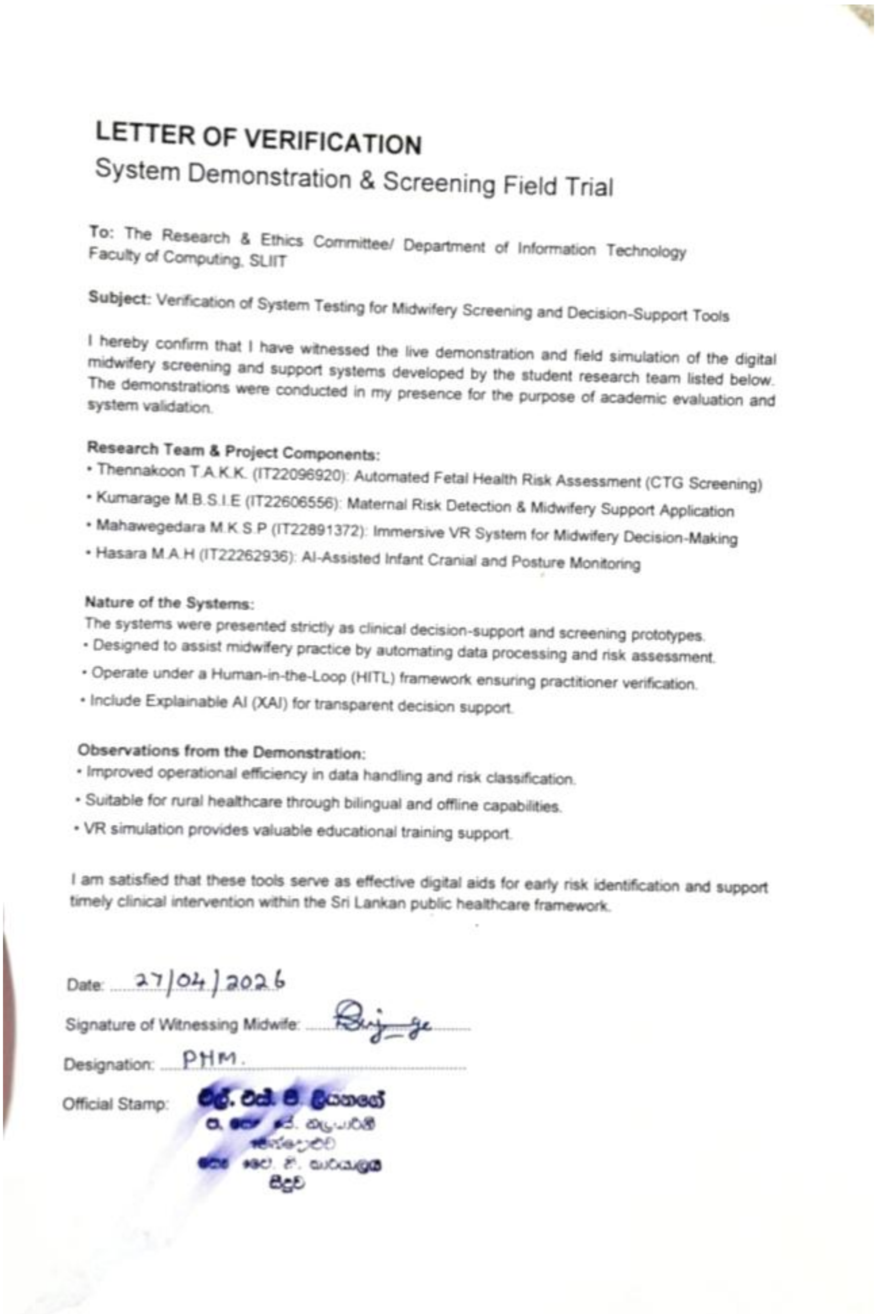
Code Segment G.7: Local saving and alert triggering logic

```
// Offline-first local saving and alert logic
await localDatabase.insert('maternal_records', {
  'patientCode': patientCode,
  'systolicBP': systolicBP,
  'diastolicBP': diastolicBP,
  'bloodSugar': bloodSugar,
  'riskLevel': riskLevel,
  'createdAt': DateTime.now().toIso8601String()
});

if (riskLevel == 'High Risk') {
  triggerHapticAlert();
  playVoiceAlert(selectedLanguage);
} else if (riskLevel == 'Mid Risk') {
  triggerWarningHaptic();
}
```

Appendix H: System Demonstration Verification Letter

This appendix includes the scanned verification letter confirming that the system demonstration and screening field trial were witnessed for academic evaluation. The letter supports the practical validation evidence discussed in the methodology and results chapters.



LETTER OF VERIFICATION

System Demonstration & Screening Field Trial

To: The Research & Ethics Committee/ Department of Information Technology
Faculty of Computing, SLIIT

Subject: Verification of System Testing for Midwifery Screening and Decision-Support Tools

I hereby confirm that I have witnessed the live demonstration and field simulation of the digital midwifery screening and support systems developed by the student research team listed below. The demonstrations were conducted in my presence for the purpose of academic evaluation and system validation.

Research Team & Project Components:

- Thennakoon T.A.K.K. (IT22096920): Automated Fetal Health Risk Assessment (CTG Screening)
- Kumarage M.B.S.I.E (IT22606556): Maternal Risk Detection & Midwifery Support Application
- Mahawegedara M.K.S.P (IT22891372): Immersive VR System for Midwifery Decision-Making
- Hasara M.A.H (IT22262936): AI-Assisted Infant Cranial and Posture Monitoring

Nature of the Systems:

The systems were presented strictly as clinical decision-support and screening prototypes.

- Designed to assist midwifery practice by automating data processing and risk assessment.
- Operate under a Human-in-the-Loop (HITL) framework ensuring practitioner verification.
- Include Explainable AI (XAI) for transparent decision support.

Observations from the Demonstration:

- Improved operational efficiency in data handling and risk classification.
- Suitable for rural healthcare through bilingual and offline capabilities.
- VR simulation provides valuable educational training support.

I am satisfied that these tools serve as effective digital aids for early risk identification and support timely clinical intervention within the Sri Lankan public healthcare framework.

Date: 27/04/2026

Signature of Witnessing Midwife:

Designation: PHM.

Official Stamp:

Appendix H Figure 1: Verification letter for system demonstration and screening field trial